

Revealing hidden Market Regimes in Bitcoin using Unsupervised Machine Learning

Semester Project to the Lecture

Unsupervised Machine Learning

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Abstract

Bitcoin is characterized by repeating bull and bear markets, periods of strong speculation, rapid growth, crashes and changing market conditions. Identifying such market regimes can help to better understand Bitcoin market behaviour and potentially improve investment and risk management decisions. The goal of this project is to investigate whether meaningful Bitcoin market regimes can be identified using Unsupervised Machine Learning techniques. For this purpose, a dataset containing market, network and on-chain variables was collected and three clustering algorithms were evaluated.

The results show that Bitcoin market regimes can be identified successfully using Unsupervised Machine Learning. One of the most important findings is that feature engineering had, with some exceptions, a larger impact on the final results than the clustering algorithm itself. While all algorithms were applied to the same underlying information, the selected feature representation significantly influenced the resulting regime structures. Dataset C based on rolling Z-Scores combined with K-Means clustering produced the most convincing results (K-Means C-k4). The identified clusters consistently captured bull markets, bear markets and transitional phases reliably while also showing surprisingly early signals of upcoming market regime changes.

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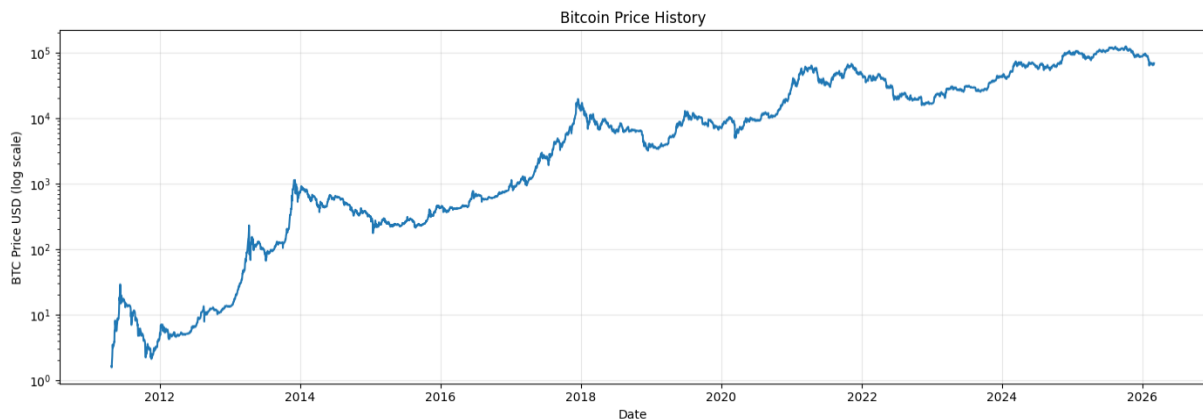
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1. Introduction

Bitcoin and other cryptocurrencies exhibit particularly strong structural market phases. Examples include bull and bear markets, accumulation and distribution periods, liquidity events and speculative peaks. While the internet is full of attempts to identify such phases using price charts and technical indicators, these approaches often ignore the large amount of additional information that public blockchains provide transparently.

The core idea of this project is therefore to move beyond price-based regime identification and instead use a broader set of market, network and on-chain variables combined with Unsupervised Machine Learning techniques to uncover hidden market structures. If such structures can be identified reliably, they may not only help explain past market behaviour but potentially also acting as early signals of upcoming regime changes and trend reversals.

Figure 1: Bitcoin Price History (Log Scale): Patterns of Recurring Ups and Downs



Source: CoinMetrics, Own Calculations

The **research question** of this project is therefore:

Can Unsupervised Machine Learning methods identify useful Bitcoin market regimes using market, network and on-chain data, that can potentially also act as early warning signals of upcoming change?

To answer this question, an initial dataset was obtained containing market variables (price, returns, volume), short- and longterm-demand on-chain metrics (active addresses, hash rate, fees, transaction activity), exchange-related metrics (exchange inflows, outflows and balances) as well as valuation measures such as MVRV. In addition, features were engineered to incorporate historical developments and longer-term market dynamics - the step that took up by far the most time, while also yielding the highest return - followed by scaling and dimensionality reduction using PCA. Finally, three different Unsupervised Machine Learning Algorithms were applied and evaluated: **K-Means**, **Gaussian Mixture Models** and **HDBSCAN**.

Based on Bitcoin's recurring market cycles, strong speculation and herding behaviour, I expected the algorithms to identify multiple distinct market regimes rather than a simple bull-versus-bear

classification. Furthermore, I initially expected HDBSCAN to perform best, as different market regimes could potentially form distinct density regions within the feature space. Regarding feature engineering, I expected Dataset B to perform best, as the longer growth horizon appeared to capture regime-like structures more effectively during the exploratory analysis.

The results showed a different picture. While multiple meaningful market regimes could indeed be identified successfully, Dataset C based on rolling Z-Scores consistently produced the strongest results. In addition, K-Means clearly outperformed the other clustering methods, while HDBSCAN struggled to identify useful regime structures. Not only did K-Means find reliable and consistent market regimes, but also possible early regime change signals up to months before the trend reversal happened.

The project follows the following workflow:

1. Explore and preprocess Bitcoin market, network and on-chain data.
2. Construct three different feature datasets using growth rates, moving averages and rolling Z-Scores.
3. Standardize the features and apply Principal Component Analysis (PCA).
4. Apply and compare K-Means, Gaussian Mixture Models (GMM) and HDBSCAN.
5. Evaluate the resulting clusters using quantitative metrics, visual inspection and economic reasoning.
6. Select and discuss the preferred model.

The remainder of this report is structured as follows:

- Chapter 2 presents the dataset and preprocessing steps.
- Chapter 3 describes the feature engineering process and construction of the different datasets.
- Chapter 4 introduces PCA and dimensionality reduction.
- Chapter 5 presents the clustering analysis and compares the different algorithms.
- Chapter 6 discusses the results and compares the different approaches.
- Chapter 7 concludes the project and outlines possible future improvements and extensions.

2. Dataset and Data Exploration

2.1 Dataset Cleaning and Preparation

The initial dataset was obtained from CoinMetrics and contained 6,269 daily observations and 32 columns covering market, network and on-chain activity of Bitcoin from early 2011 until March 2026.

Before starting the actual analysis, the dataset was cleaned and reduced based on domain knowledge and the objective of the project. The goal is not to cluster observations according to calendar time, but to identify recurring Bitcoin market regimes such as bull markets, bear markets and transitional phases.

This distinction is particularly important because many Bitcoin-related variables contain strong long-term trends caused by the growth and adoption of the network. Such variables are not necessarily problematic, as later feature engineering steps can remove these trends through growth rates, moving averages or rolling Z-Scores. Variables were therefore not removed simply because they contained trends.

However, variables that mainly represent deterministic time progression were treated differently. A prominent example is Bitcoin supply. Since supply follows a predetermined issuance schedule and grows monotonically over time, it primarily contains information about where an observation lies within Bitcoin's history rather than information about the current market regime. Such variables were therefore excluded from the analysis.

Based on economic reasoning and domain knowledge, an initial variable selection was performed. Variables without clear economic relevance for regime detection were removed, while potentially useful variables were retained for further exploration. Several variables were also marked for later reassessment due to possible redundancy. This step already reduced the dataset from 31 ('date' is a column) to 15 variables.

The next steps involved cleaning the remaining set by removing observations that were always concentrated entirely in the early Bitcoin years

2.2 Missing Values

A first inspection revealed that several variables showed 561 missing observations at the beginning of the sample. Investigation revealed that these missing values were concentrated entirely in the early Bitcoin years, where market and exchange-related information was not yet available. Since all affected variables became available simultaneously, the corresponding early observations were removed from the dataset.

The BTC/ETH exchange rate was also examined as a potential indicator of Bitcoin's relative strength within the broader cryptocurrency market. However, the variable contained only six valid observations and was therefore excluded due to insufficient data coverage, leaving 14 variables.

Finally, several exchange-related variables contained long sequences of zero values at the beginning of the sample. These values were interpreted as unavailable data rather than actual observations and were removed accordingly. After these adjustments, no problematic missing values remained in the dataset.

The resulting cleaned dataset serves as the foundation for the exploratory analysis and subsequent feature engineering steps.

2.3 Exploratory Data Analysis and Feature Engineering Decisions

The exploratory analysis was performed using both histograms and time-series plots for all remaining variables. While histograms provide information about distributions, skewness and extreme values, time-series plots allow assessment of long-term trends, regime-like behaviour and possible structural breaks.

Since the objective of this project is to identify Bitcoin market regimes rather than simply separate early and late observations, the temporal behaviour of each variable is particularly important. Variables with strong deterministic trends can cause clustering algorithms to group observations based on time rather than underlying market conditions, and will need a respective transformation in order to handle such trends.

Figure 2: Histogram and timeseries inspection of active_addresses

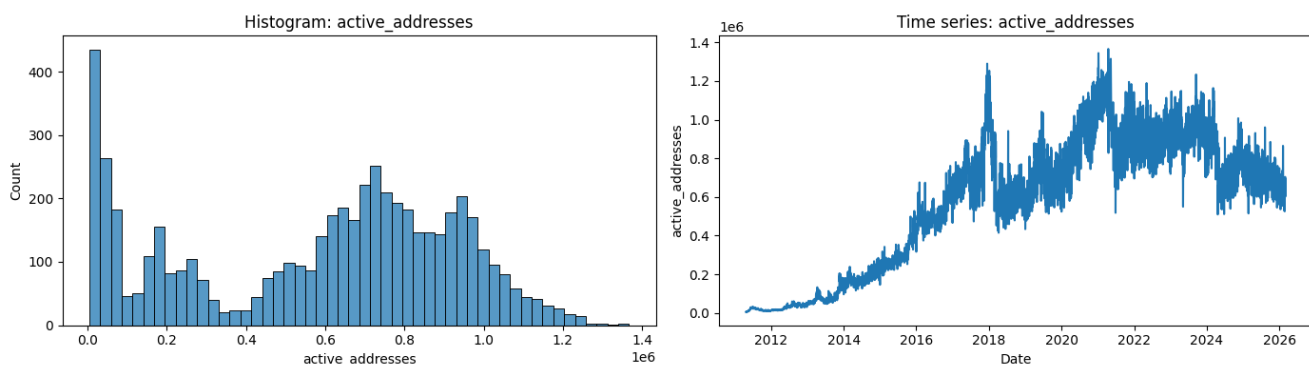


Figure 2 illustrates this process using active addresses as an example. The histogram suggests a highly skewed distribution, while the time-series plot reveals that this skewness is largely driven by Bitcoin's early adoption phase. After approximately 2016, the variable becomes substantially more stable and starts exhibiting fluctuations that could be associated with different market regimes.

While long-term trends can often be removed through later transformations such as growth rates or rolling Z-Scores, the period before 2016 represents a different type of problem. During Bitcoin's early adoption phase, many variables experienced a structural growth process that differs fundamentally from the behaviour observed afterwards. Even after transforming the variables into growth rates, this period would still contain systematically higher growth than later years and would therefore continue to encourage clustering into "early Bitcoin" versus "later Bitcoin" instead of actual market regimes. For this reason, observations before 2016 would be removed for this case whenever appropriate.

Example conclusion (active addresses):

- Remove observations before 2016.
- Transform the variable using either growth rates or rolling Z-Scores.
- No moving average smoothing appears necessary due to the moderate noise level.

The same procedure was applied to all remaining variables. Instead of discussing each variable individually, Table 1 summarizes the final decisions resulting from the exploratory analysis, while the notebooks contain all comparisons, discussions and conclusions.

Before recommending the final feature transformations, several possible transformations are considered:

- **Growth Rates (%-Changes):** Long-term growth rates of 30, 90 and 180 days are preferred over daily changes. Market regimes are not driven by short-term fluctuations but by larger structural shifts that occur over longer periods of time.
- **Moving Average Enhanced Growth Rates:** For highly volatile variables, growth rates can be calculated based on moving averages rather than raw observations. This reduces noise and allows the algorithms to focus on the underlying structure instead of daily randomness.
- **Rolling Z-Scores:** Rolling Z-Scores normalize variables relative to their recent history. Instead of measuring how much a variable changed, they measure how unusual the current observation is relative to recent behaviour. This may be particularly useful for identifying regime changes.

Two transformations were considered but ultimately not used:

- **Logarithmic Transformations:** While logarithms reduce skewness and make growth rates more symmetric, they also compress extreme values. Since extreme values may contain exactly the information needed to identify market peaks, crashes and regime changes, logarithmic transformations are not applied.
- **Time Encoding (Sine/Cosine):** The goal is to identify regimes based on economic and on-chain information rather than calendar time. Furthermore, Bitcoin does not exhibit strong weekly, monthly or seasonal patterns that would justify cyclical time encoding. The relevant time information should already be reflected in the selected features after transformation.

Overall, the exploration confirmed that most variables contain economically meaningful information for Bitcoin regime detection. However, many variables are dominated by long-term adoption and growth trends and are therefore not directly suitable for clustering in their original form.

Most variables will therefore be transformed using growth rates or rolling Z-Scores. These transformations aim to remove long-term trends while preserving the economically relevant regime information. Only a small number of variables, most notably MVRV and total transaction fees, appeared to contain useful level information and were therefore retained in their original form.

The analysis also revealed substantial overlap between several variables. Potential redundancies will be addressed later through correlation analysis and feature selection.

Table 1 summarizes the final transformation and selection decisions that will be used in the feature engineering stage.

Table 1: Summary of the recommended Decisions for Transformations

Variable	Final Decision	Final Transformation	Time Restriction	Notes
active_addresses	KEEP	Growth Rate, Rolling Z-Score	Exclude pre-2016	Captures network activity and adoption trends
nonzero_addresses	KEEP	Growth Rate, Rolling Z-Score	Exclude pre-2016	Measures broader network adoption
block_count	DELETE	-	-	Mostly protocol randomness with limited economic meaning
transaction_count	KEEP	MA-Enhanced Growth Rate, Rolling Z-Score	Exclude pre-2016	Captures transaction activity and network usage
transfer_count	KEEP	MA-Enhanced Growth Rate, Rolling Z-Score	Exclude pre-2016	Captures transfer activity and network usage
hash_rate	KEEP	MA-Enhanced Growth Rate, Rolling Z-Score	Exclude pre-2016	Represents network security and mining activity
total_fees_btc	KEEP	Level	Exclude pre-2016	Extreme values may contain regime information
issuance_usd	KEEP	MA-Enhanced Growth Rate, Rolling Z-Score	Exclude pre-2016	Captures mining reward value and possible level-shift signals
exchange_supply_btc	KEEP	MA-Enhanced Growth Rate, Rolling Z-Score	Exclude pre-2016	Measures Bitcoin held on exchanges
exchange_inflow_usd	DELETE	-	Exclude pre-2016	Correlation > 0.95 with exchange outflows
exchange_outflow_usd	KEEP	MA-Enhanced Growth Rate, Rolling Z-Score	Exclude pre-2016	Captures exchange activity and market participation
price_usd	KEEP	MA-Enhanced Growth Rate, Rolling Z-Score	Exclude pre-2016	Core market variable
mrv_ratio	KEEP	Level	Exclude pre-2016	Important valuation metric, preserved in original form
spot_volume_usd	KEEP	MA-Enhanced Growth Rate, Rolling Z-Score	Exclude pre-2016	Measures trading activity and speculation

3. Preprocessing and Feature Engineering

3.1 Feature Engineering Strategy

The exploratory analysis showed that most variables contain useful information for regime detection, but also that the majority are dominated by long-term trends and short-term fluctuations. The objective of the feature engineering stage is therefore to remove these unwanted effects while preserving the information that may characterize different Bitcoin market regimes.

A central question is the appropriate time horizon for the transformations. Market regimes are typically not driven by daily fluctuations but by larger structural changes that develop over months. Therefore, the transformation window must be long enough to capture these developments while still reacting sufficiently fast to regime changes.

Since there is no universally correct answer and the optimal choice is ultimately data-dependent, I decided to compare several reasonable approaches instead of relying on a single specification. The goal is to evaluate which representation of the same underlying information allows the clustering algorithms to identify the most meaningful market regimes.

Three transformation methods are considered:

- **Growth Rates:** Instead of daily percentage changes, longer-term growth rates are calculated. This captures structural changes rather than short-term noise. Based on the exploratory analysis, 90-day and 180-day horizons appear most suitable.
- **Moving Average Enhanced Growth Rates:** Many variables exhibit high daily volatility. To reduce the influence of random fluctuations, 30-day moving averages are calculated first and the growth rates are then derived from these smoothed values. This preserves the medium- and long-term structure while reducing noise.
- **Rolling Z-Scores:** Rolling Z-Scores measure how unusual the current observation is relative to its recent history. Instead of focusing on absolute changes, this approach highlights deviations from normal behaviour and may therefore be especially useful for identifying regime changes and market extremes.

The same transformation length is applied to all variables requiring transformation. While it would be possible to optimize each variable individually, such fine-tuning would introduce additional complexity without any evidence that it improves the results. A consistent transformation framework also makes the resulting datasets easier to interpret and compare.

Observations before 2016 are removed prior to feature construction. As discussed in the exploratory analysis, the early Bitcoin years represent a fundamentally different adoption phase that would otherwise influence the transformed features. Removing this period before constructing the features ensures that only observations from the relevant market environment contribute to the final variables.

Importantly, the variables **MVRV Ratio** and **Total Fees BTC** are retained in levels and are not transformed. The exploratory analysis suggested that their absolute values may themselves contain economically meaningful information regarding valuation and market extremes.

3.2 Final Feature Sets

To evaluate different representations of Bitcoin market behaviour, three separate datasets are constructed:

Feature Set A: 90-Day Growth Rates

- 90-day growth rates calculated from 30-day moving averages
- Captures medium-term structural changes
- Retains relatively high responsiveness to regime transitions

Feature Set B: 180-Day Growth Rates

- 180-day growth rates calculated from 30-day moving averages
- Captures longer-term market developments
- Expected to produce smoother regime structures

Feature Set C: Rolling Z-Scores (365 Days)

- 365-day rolling Z-Scores
- Measures how unusual current observations are relative to the previous year
- Focuses on deviations from normal market behaviour rather than absolute growth

The resulting datasets represent three different views of the same underlying Bitcoin market information. In the following sections, these feature sets will be validated, reduced where necessary and prepared for the clustering analysis.

3.3 Feature Validation

Before proceeding to dimensionality reduction and clustering, the engineered features are validated once more. The objective of this step is to verify that the transformations achieved their intended purpose while preserving meaningful information.

For each variable, the raw version is compared against the transformed versions of Dataset A, Dataset B and Dataset C using both histograms and time-series plots.

The key questions are:

Histograms

- Are the distributions plausible?
- Is meaningful structure still present?
- Are there any unrealistic values or artefacts?

Time-Series Plots

- Have the long-term trends been removed?
- Can potential regime structures still be observed?
- Were the variables over-smoothed through the transformations?

3.3.1 Histogram Comparison

The results are generally very encouraging. As Figure 3 shows, across almost all variables, the transformations substantially reduced skewness and produced distributions that are considerably closer to normality.

At the same time, the transformations did not remove all variation. The resulting distributions still exhibit visible structure rather than collapsing into random noise. This is important because the objective is not simply to normalize the variables, but to preserve the information that may characterize different market regimes.

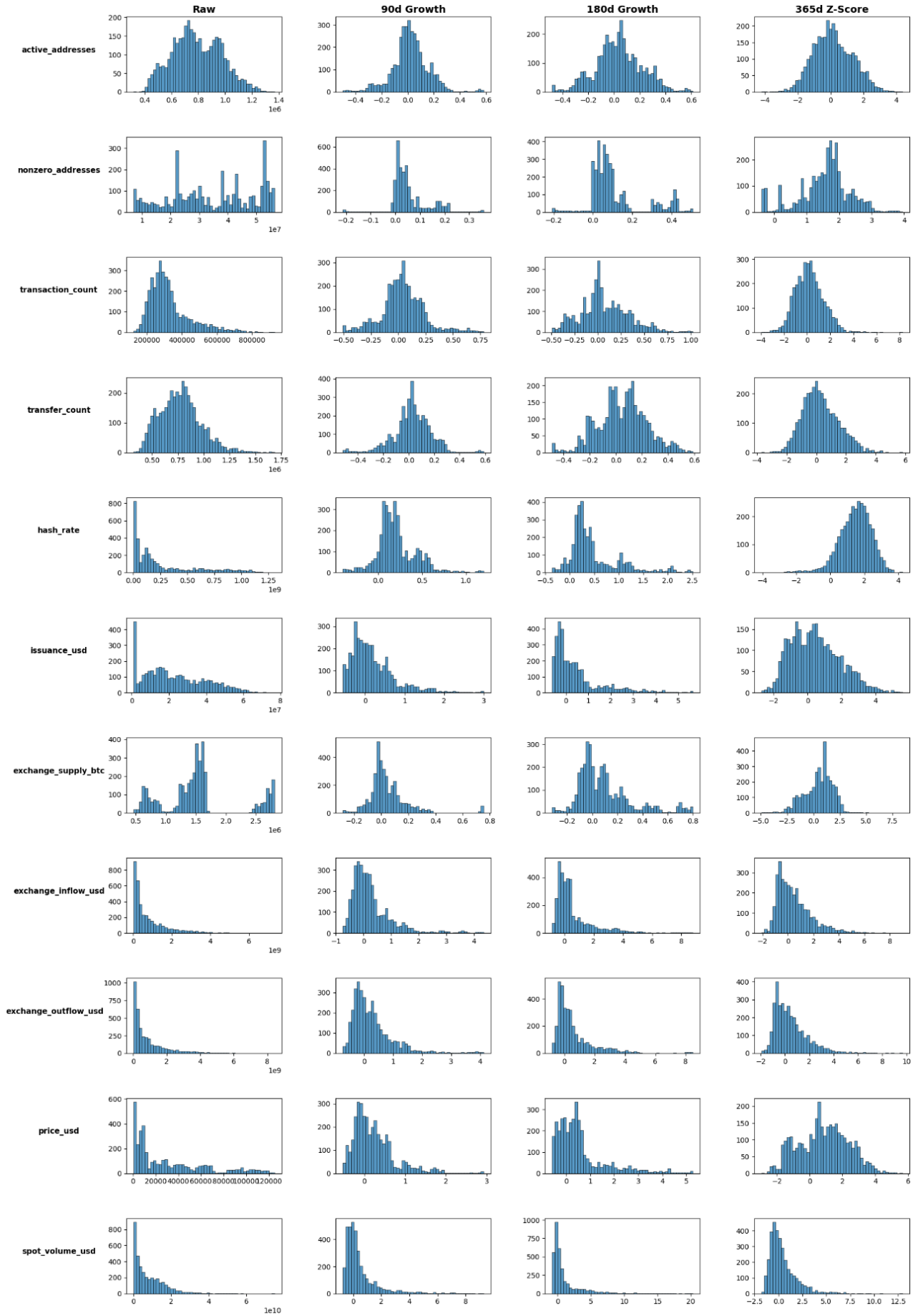
Interestingly, neither Dataset A nor Dataset B appears universally superior from a purely distributional perspective. Some variables look more balanced under the 90-day growth specification, while others appear more symmetric under the 180-day growth specification.

Dataset C generally produces the most regular distributions. The rolling Z-Score transformation successfully reduces extreme skewness while maintaining variation across the full range of observations.

As expected, MVRV and Total Fees remain unchanged since they were intentionally retained in levels.

Overall, the histogram analysis suggests that all three feature engineering approaches produce plausible and economically meaningful variables without introducing obvious distortions.

Figure 3: Histogram Comparison of Raw Dataset vs Transformed Datasets A, B, C



3.3.2 Time-Series Comparison

The time-series comparison, shown by Figure 4 is more important than the histogram analysis because the main objective of this project is to identify market regimes rather than achieve specific statistical distributions.

The most important result is that the strong long-term trends visible in the raw data have largely disappeared after transformation. Most transformed variables fluctuate around relatively stable levels and therefore exhibit approximate stationarity.

The 90-day growth specification remains somewhat more volatile and responsive, while the 180-day growth specification appears smoother and focuses more strongly on medium-term developments. Both transformations remove the original trends very effectively while preserving clear periods of expansion, contraction and unusual market behaviour.

The rolling Z-Score approach produces a similar overall structure while often appearing even more stationary. At the same time, several variables show higher short-term volatility than under the growth-rate approaches. This is expected, as no moving-average smoothing was applied prior to the Z-Score calculation.

Most importantly, the transformed variables still exhibit visible regime-like structures. Bull market periods, bear markets and transitional phases often remain clearly identifiable even after the transformations. This indicates that the transformations removed the long-term trend while preserving economically meaningful information.

One remaining observation is that some growth-rate variables still appear slightly elevated during the 2016-2018 period compared to later years. This may reflect Bitcoin's early adoption phase and diminishing-return dynamics. Whether this affects clustering quality cannot be determined from visual inspection alone and will therefore be evaluated during the clustering stage.

Finally, Total Fees remains the most unusual variable. While it contains potentially valuable information about market extremes and network congestion, the extreme spikes may also influence distance-based clustering algorithms disproportionately. Depending on the clustering results, this variable may become a candidate for removal in future iterations.

3.3.3 Conclusion

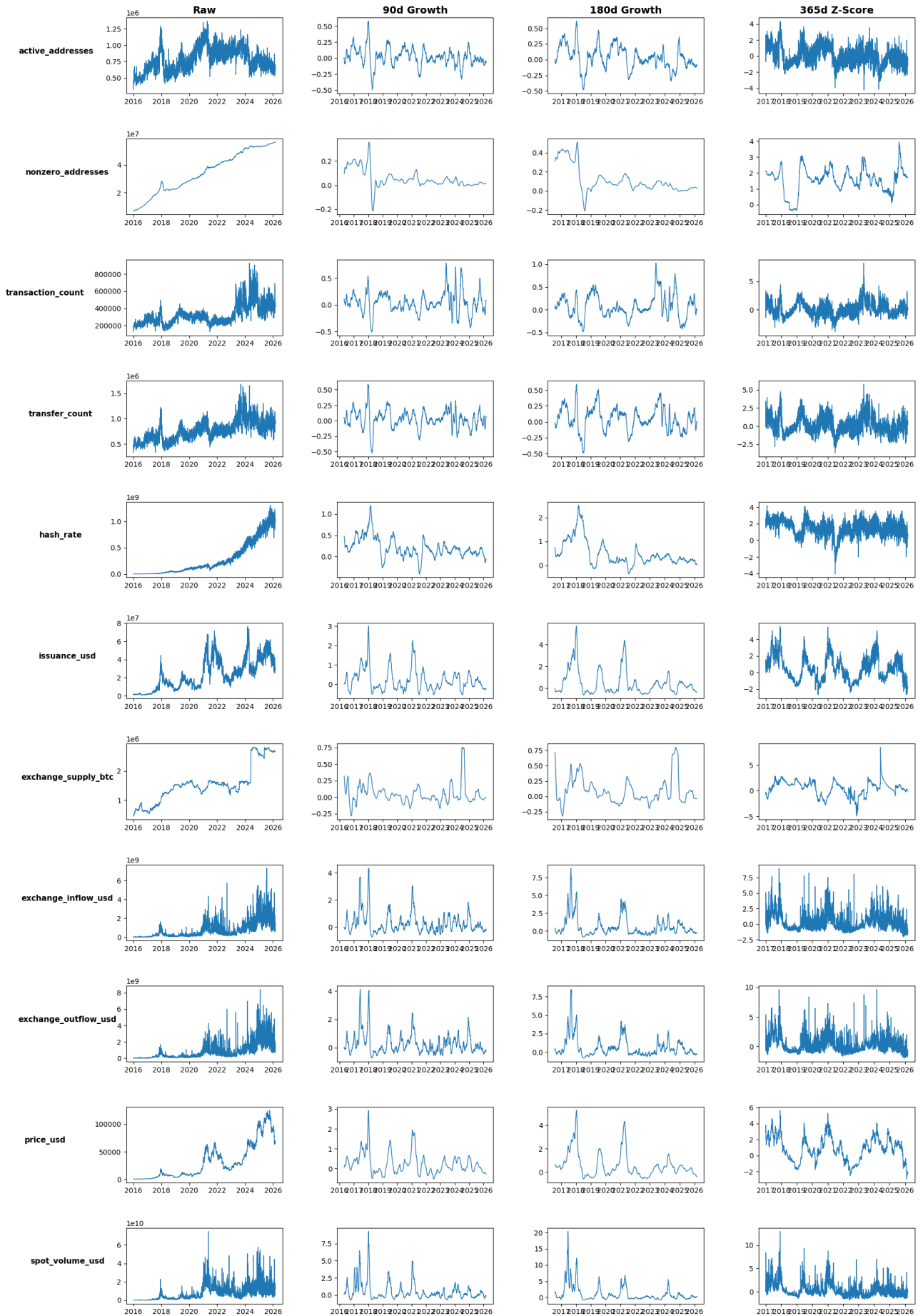
The feature validation confirms that the engineering process achieved its primary objective. Long-term trends were largely removed while economically meaningful regime structures remained visible.

All three feature sets appear suitable for clustering and no major issues were identified that would justify revisiting the transformations at this stage.

A few potential improvements remain possible. The rolling Z-Score approach could potentially benefit from additional smoothing, the early 2016-2018 period may still influence some growth-rate variables, and the extreme behaviour of Total Fees may affect specific clustering algorithms. However, these questions can only be answered reliably after evaluating the clustering results themselves.

The engineered features are therefore approved and the analysis proceeds to feature selection.

Figure 4: Timeseries Comparison of Raw Dataset vs Transformed Datasets A, B, C



3.4 Feature Selection

3.4.1 Selection Strategy

The feature selection step can have a large impact on the results of Unsupervised Machine Learning. Unlike in Supervised Machine Learning, the objective is not to maximize predictive performance, but rather to preserve the underlying structure of the data while avoiding redundant information and, especially in this project, avoiding an overrepresentation of specific economic concepts.

The first objective is therefore the elimination of redundant information, meaning information that is already represented by another variable. To identify such cases, correlation matrices are used as a diagnostic tool. Variables with correlations above 0.95 are considered to contain nearly identical information and are candidates for removal.

The second objective is more subtle and concerns the overrepresentation of information. Even if pairwise correlations are not extremely high, repeatedly measuring the same underlying concept can implicitly increase its weight within the dataset and therefore influence the clustering results.

This becomes particularly important once dimensionality reduction is applied. Both linear methods such as PCA and non-linear methods such as UMAP tend to assign greater importance to information that appears repeatedly throughout the dataset. As an example, consider a dataset with ten variables where eight primarily measure trading demand. Even if all variables contain useful information, the reduced representation will largely be driven by demand-related signals while assigning less weight to other potentially important dimensions.

For Bitcoin, the variables can be grouped into the following primary economic dimensions:

Primary Dimension Measurement	Variables
Demand & Short-Term Usage	active_addresses, transaction_count, transfer_count, spot_volume_usd
Long-Term Demand	nonzero_addresses
Valuation	mvr_ratio, price_usd
Supply Economics / Valuation	issuance_usd
Mining	hash_rate
Exchange Behaviour	exchange_supply_btc, exchange_inflow_usd, exchange_outflow_usd
Network Stress	total_fees_btc

Active Addresses, Transaction Count, Transfer Count, Spot Trading Volume and, to some extent, Nonzero Addresses all primarily measure demand-related activity. If all variables were assumed to contribute equally, demand would therefore become the dominant signal within the dataset. This is not necessarily wrong. If demand is indeed the most important driver of Bitcoin market regimes, then such weighting would be justified. However, the dataset should ideally represent multiple dimensions that may contribute to regime formation rather than being dominated by a single economic concept.

Consequently, feature selection is not only about removing redundant variables, but also about ensuring that the dataset as a whole captures a broad range of economically relevant dimensions. The final decisions will therefore be guided by both correlation patterns and economic reasoning.

The overall goal is not to maximize the number of variables, but to construct a compact and economically diverse representation of Bitcoin market conditions that allows the clustering algorithms to identify meaningful market regimes rather than merely reflecting one dominant aspect of the system.

To achieve this, a two-step selection procedure is applied:

1. Variables with correlations above 0.95 are removed. The decision regarding which variable to remove is based on economic interpretation as well as on which variable exhibits stronger correlations with the remainder of the dataset.
2. Dimensions that appear overrepresented are evaluated separately. Variables within the same economic dimension showing correlations above 0.80 are reviewed and may be removed if they contribute limited additional information. A correlation above 0.80 alone does not automatically imply elimination. The primary criterion remains economic reasoning.

At this stage, demand appears to be a reasonable candidate for receiving a relatively high weight and therefore no reduction is planned within this dimension. However, `issuance_usd` may contribute limited new information beyond valuation-related variables, while exchange behaviour currently receives a comparatively large weight through `exchange_supply_btc`, `exchange_inflow_usd` and `exchange_outflow_usd`. These dimensions will therefore receive particular attention during the selection process.

The procedure is applied separately to each dataset, as the different feature transformations result in different correlation structures. For each dataset, a correlation matrix will be analyzed and all correlations above 0.80 will be inspected in detail.

3.4.2 Selection Results

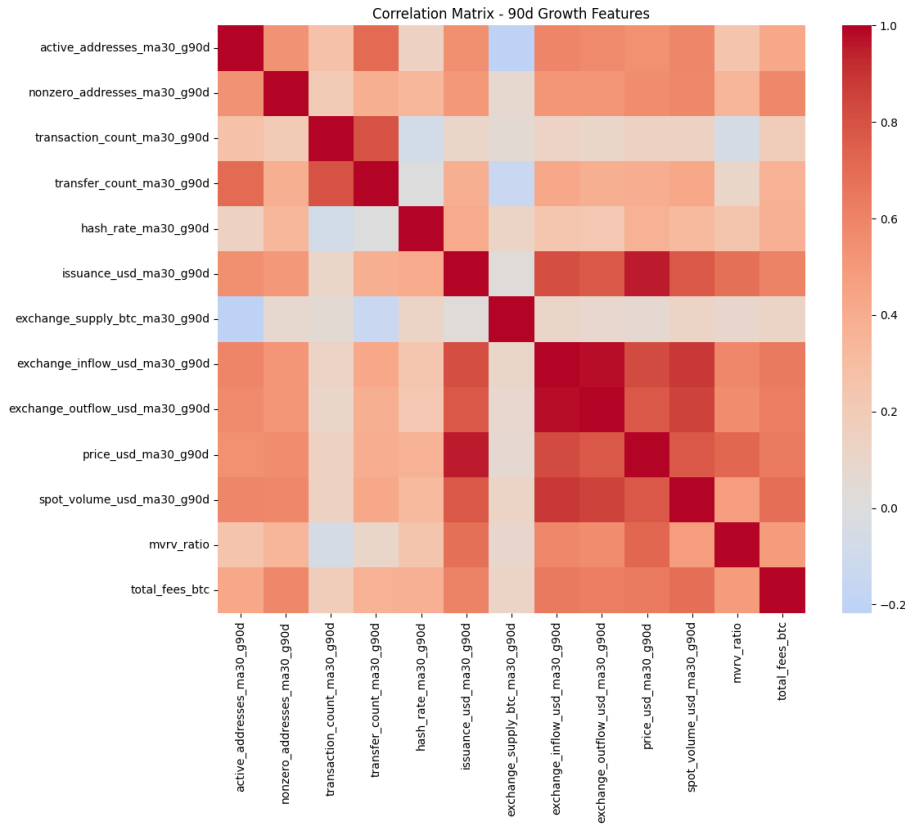
Dataset A

Figure 5 shows the correlation matrix for Dataset A and gives a first overview of how much overlap exists between the different variables.

Several correlations above the threshold of 0.80 can be observed, as indicated by Table 2. The most obvious one is between `exchange_inflow_usd` and `exchange_outflow_usd` with a correlation of 0.977. Economically this is not surprising, as periods with large inflows are often also periods with large outflows. The second very high correlation appears between `issuance_usd` and `price_usd` at 0.956. This is reasonable as `issuance_usd` is directly affected by Bitcoin's price level.

Besides these two cases, several other correlations above 0.80 appear as well. Transaction Count and Transfer Count show a correlation of 0.803, while Exchange Flows, Spot Volume and Price are all relatively closely connected. However, these variables still represent somewhat different economic concepts and therefore are not automatically candidates for removal.

Figure 5: Correlation Matrix of Dataset A



Based on the predefined threshold of 0.95, two variables are removed.

- The first one is **exchange_inflow_usd**. While inflows and outflows contain nearly identical information, inflows generally correlate more strongly with the rest of the dataset. Therefore, exchange_outflow_usd is retained as the more distinct variable.
- The second variable removed is **issuance_usd**. The variable is highly correlated with price_usd while at the same time primarily representing Bitcoin's deterministic issuance schedule multiplied by the current price. It therefore adds relatively little information while increasing the weight of valuation-related information.

No additional variables are removed. Although some demand-related variables show moderately high correlations, they still measure different aspects of network activity and adoption.

Table 2: Correlation >0.80 in Dataset A

Variable 1	Variable 2	Correlation
exchange_inflow_usd_ma30_g90d	exchange_outflow_usd_ma30_g90d	0.976737
issuance_usd_ma30_g90d	price_usd_ma30_g90d	0.955812
exchange_inflow_usd_ma30_g90d	spot_volume_usd_ma30_g90d	0.884495
exchange_outflow_usd_ma30_g90d	spot_volume_usd_ma30_g90d	0.854473
exchange_inflow_usd_ma30_g90d	price_usd_ma30_g90d	0.820382

Dataset B

The correlation structure of Dataset B looks extremely similar to Dataset A. This is not surprising, as both datasets are built from the same underlying variables and only differ in the selected growth horizon.

Again, the strongest relationships appear between `exchange_inflow_usd` and `exchange_outflow_usd` (0.989) as well as `issuance_usd` and `price_usd` (0.971). In fact, both correlations are even slightly stronger than in Dataset A. The remaining correlations above 0.80 are also largely concentrated around valuation and exchange-related variables. Overall, the picture is therefore almost identical to the previous dataset. As a consequence, the same variables are removed.

`exchange_inflow_usd` is removed because it contains almost identical information to `exchange_outflow_usd`, while `issuance_usd` is removed because it strongly overlaps with `price_usd` and contributes limited new information. No further reductions appear necessary.

Dataset C

Dataset C produces a noticeably different correlation structure. While several correlations above 0.80 still exist, none of them exceed the redundancy threshold of 0.95.

The strongest correlation is observed between `exchange_inflow_usd` and `exchange_outflow_usd` at 0.898. Other relatively high correlations appear between `price_usd` and `mrv_ratio` (0.859), `transaction_count` and `transfer_count` (0.856), as well as `issuance_usd` and `price_usd` (0.838).

From a purely statistical perspective, no variable would have to be removed here due to redundancy. However, the second objective of this chapter is not only redundancy reduction but also avoiding overrepresentation of specific dimensions.

In Dataset C, exchange behaviour is still represented by three separate variables: `exchange_supply_btc`, `exchange_inflow_usd` and `exchange_outflow_usd`. Since inflows and outflows are still highly correlated and exchange behaviour already receives a relatively large weight in the dataset, I decided to remove `exchange_inflow_usd`. All remaining variables are retained. While some correlations are still relatively high, they represent economically different concepts and therefore add useful information to the dataset.

3.5 Feature Scaling

Before applying dimensionality reduction and clustering methods, all features need to be scaled. This step is necessary because every dimensionality reduction and clustering method used in this project is based on distances between observations. Without scaling, variables with larger numerical ranges would automatically receive a higher weight and dominate the resulting principal components and clusters, regardless of their actual importance for identifying market regimes.

To avoid this issue, all variables are standardized using the **StandardScaler** transformation. **StandardScaler** transforms each variable to have a mean of zero and a standard deviation of one. This ensures that all features contribute equally to the analysis while preserving the underlying structure and information contained in the data. The scaling is performed separately for each

feature set after feature selection. The resulting scaled datasets are then used as the input for dimensionality reduction and the subsequent clustering analysis.

4. Dimensionality Reduction

4.1 Overview

The feature engineering process resulted in multiple transformed and scaled feature sets containing different aspects of Bitcoin market activity, valuation, mining, exchange flows and network demand.

Before starting with clustering and trying to find different regimes in Bitcoin time series, it is useful to first reduce the dimensionality of the data due to two reasons:

1) Overlapping information: Dimensionality Reduction methods do not remove overlapping information, but represent it in a lower-dimensional space, basically summarizing it into a smaller number of components. This reduces redundancy and noise while preserving most of the underlying information in the dataset.

2) Mitigating the 'Curse of Dimensionality': The more dimensions a dataset has, the more possibilities each datapoint has to be further away from others. This results in higher distances, less close neighbours and generally more datapoints in solitude the higher the dimensions and generally leads to distorted results in clusters, or even with the possibility of not finding any clusters at all. For the case of my datasets, although the number of features is relatively small, dimensionality reduction helps reduce potential effects of the curse of dimensionality and improves the representation of the underlying data structure.

For Dimensionality Reduction two different approaches exist:

- Linear dimensionality reduction (e.g. PCA)
- Non-linear dimensionality reduction (e.g. UMAP, t-SNE)

Principal Component Analysis (PCA) transforms the original features into a smaller set of orthogonal components that explain as much variance in the data as possible. PCA is also highly interpretable and allows analysis of the underlying economic factors driving the dataset.

Non-linear methods aim to preserve the structure of the data in a lower-dimensional space without assuming linear relationships between variables. These methods can capture more complex patterns that may not be visible with PCA, and therefore could lead to potentially different results in clustering.

In this project, PCA is applied to all feature sets and serves as the primary dimensionality reduction technique. Clustering algorithms are first evaluated on the PCA representations of all datasets. After selecting the best performing model, t-SNE is used as a visualization tool to provide an intuitive two-dimensional representation of the discovered clusters.

The next steps for the dimensionality reduction with PCA are as follows:

1. Apply PCA to each feature set.
2. Analyze the explained variance and determine the appropriate number of principal components.
3. Explore the PCA loadings to understand which variables drive the resulting components.
4. Transform each dataset into its lower-dimensional PCA representation.

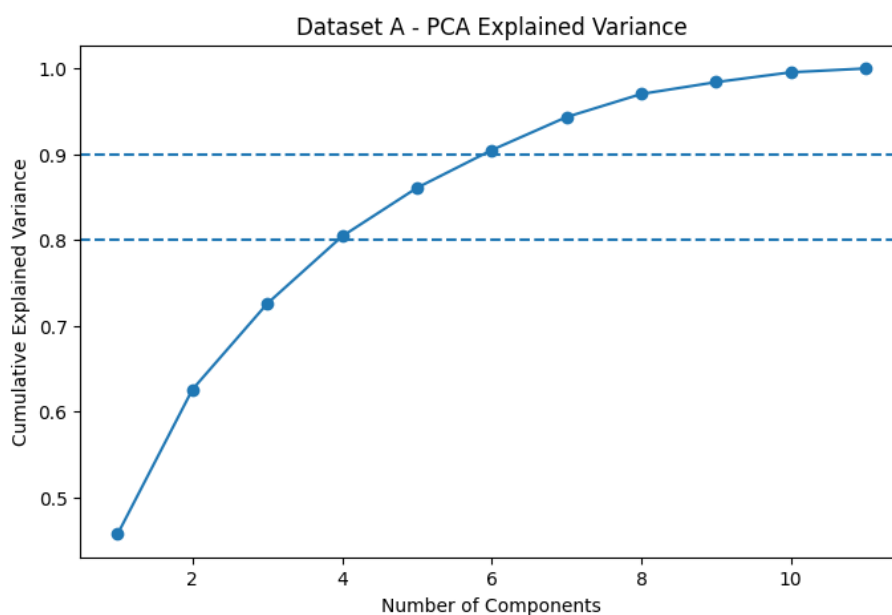
5. After these steps are taken, the PCA-transformed datasets are ready for the clustering algorithms.

4.2 PCA Dataset A

Interpretation

Principal Components are new variables created as linear combinations of the original features. They represent the main directions of variation in the dataset, with each component capturing a different source of information. The loadings (see *Appendix A* for numerical details) indicate how strongly each original feature contributes to a given component and can therefore be used to interpret its economic meaning.

Figure 6: PCA Scree Plot of Dataset A: Explained Variance by Number of Components



The first principal component explains 45.8% of the total variance and is mainly driven by price growth, spot volume growth, exchange outflows, total fees and address activity. This suggests that the dominant source of variation in the dataset is related to overall trading activity and speculative demand, what is very reasonable given that it can be classified as a speculative asset.

The second principal component is primarily influenced by transaction count and transfer count, while showing an opposite relationship with valuation and mining-related variables. It possibly captures changes in transaction activity that are partly independent from overall market performance, probably showcasing the variation through usage-demand instead of pure trading-and speculation.

The third principal component is strongly associated with exchange supply, indicating that exchange balances contain information that is not fully captured by the other variables.

The fourth principal component is dominated by hash rate growth and can be interpreted as a mining-related factor.

In total, the first 4 principal components explain approximately 80% of the total variance, while 6 components explain more than 90%.

Overall, the PCA results suggest that the selected features capture multiple economic dimensions of the Bitcoin ecosystem instead of only reflecting a single demand or price factor.

Conclusion

6 components, explaining 90.5% of the total variance in the dataset, will be selected for the clustering analysis. This decision provides a good balance between dimensionality reduction and information retention. While 4 components already explain approximately 80% of the variance, the additional two components increase the explained variance to over 90% and therefore keep a substantially larger share of the original structure of the data.

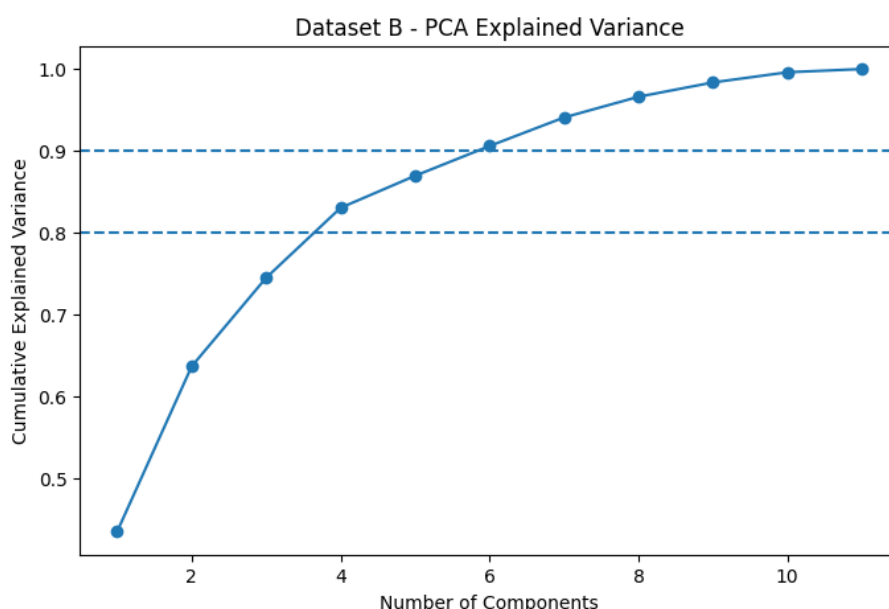
The PCA results also indicate that the feature engineering process was successful in preserving different sources of information, including market activity, exchange behaviour and mining dynamics.

4.3 PCA Dataset B

Interpretation

Generally, Dataset B produces very similar principal component structures to Dataset A, indicating that the results are robust across different growth time horizons. Dataset B showed a slightly more balanced distribution of explained variance between the principal components, suggesting that the longer growth horizon may capture economic regimes a bit more effectively.

Figure 7: PCA Scree Plot of Dataset B: Explained Variance by Number of Components



The principal components of Dataset B are analogous to the ones of A primarily driven by the same variables (see *Appendix A*). In total, the first four principal components explain approximately 83.0% of the total variance, while six components explain more than 90%.

Overall, the PCA results suggest that the selected features capture several distinct economic dimensions of the Bitcoin ecosystem, including market activity, transaction behaviour, exchange dynamics and mining activity.

Conclusion

As the results are very similar to the ones within Dataset A, 6 principal components, explaining 90.6% of the total variance in the dataset, will too be selected for the clustering analysis for the same reason as well as to enable direct comparison of the results.

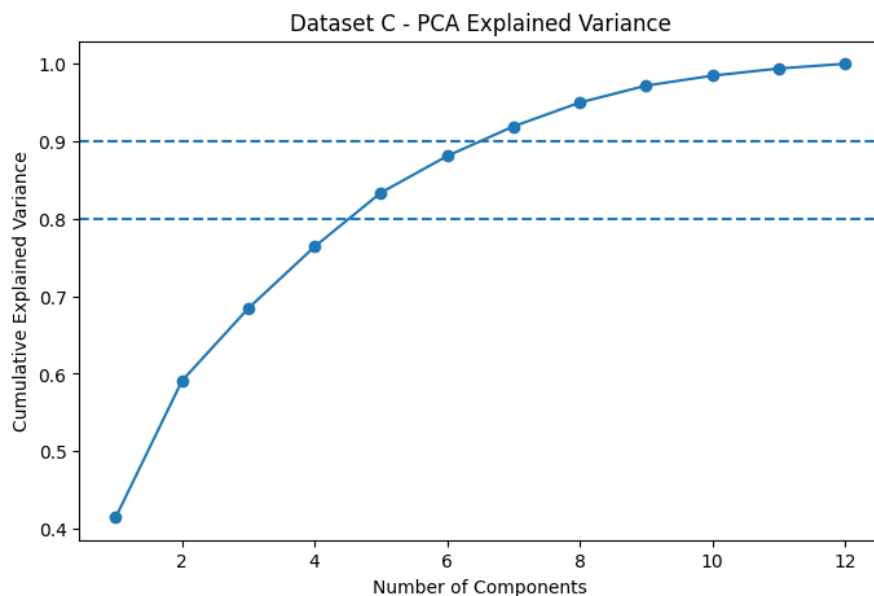
Dataset A and Dataset B produced highly similar principal component structures, indicating that the results are robust across different growth horizons. Dataset B showed a slightly more balanced distribution of explained variance, suggesting that the longer growth horizon may capture economic regimes more effectively. I will test this hypothesis with the clustering.

4.4 PCA Dataset C

PCA Interpretation

Dataset C also produces a principal component structure that is broadly similar to Dataset A and Dataset B. The main economic dimensions identified previously, including market activity, transaction behaviour, exchange dynamics and mining activity, are still clearly visible in the first principal components.

Figure 8: PCA Scree Plot of Dataset C: Explained Variance by Number of Components



However, compared to the growth-based datasets, the explained variance is distributed more evenly across the components. The first principal component explains only 41.5% of the total variance, compared to 45.8% and 43.4% in Dataset A and Dataset B respectively. As a result, more principal components are required to capture the same amount of information.

This behaviour is expected, as rolling Z-scores measure deviations from historical norms rather than long-term growth trends, while also compared to the Z-scores the growth rates were based

on a smoothed Moving-Average transformation prior. As a result, variables such as fees, issuance value and exchange activity that are more volatile in their raw form contain more unique information in the Dataset C and are less absorbed into a single dominant market activity factor.

In total, the first five principal components explain approximately 83.4% of the total variance, while seven components explain more than 90%.

Overall, the PCA results suggest that the rolling Z-score transformation captures a broader set of abnormal market conditions and distributes information more evenly across different economic dimensions of the Bitcoin ecosystem.

Conclusion

In contrast to Dataset A and Dataset B, Dataset C requires seven principal components to exceed the 90% explained variance threshold. Therefore, seven components, explaining 91.9% of the total variance, will be selected for the clustering analysis.

While the overall principal component structure remains similar to the growth-based datasets, the rolling Z-score transformation appears to preserve more variable-specific information and focuses less on a single dominant factor. This suggests that Dataset C may be especially useful for identifying unusual market conditions and abnormal regime behaviour.

The effectiveness of this alternative representation will be evaluated during the clustering analysis and compared directly with the results of Dataset A and Dataset B.

4.5 PCA Results Comparison and Conclusion

The PCA results show a very similar structure across all three datasets. In all cases, the first four principal components capture information related to market activity, transaction behaviour, exchange dynamics and mining activity. This suggests that the main economic dimensions of the Bitcoin ecosystem are preserved regardless of the chosen feature transformation.

Table 3: Comparison of PCA Results per Dataset

Dataset	Transformation	Variance Explained by PC1	Variance Explained by First 4 PCs	Variance Explained by First 6 PCs	Components required for >90% Variance
Dataset A	90-Day Growth Rate	45.8%	80.4%	90.5%	6
Dataset B	180-Day Growth Rate	43.4%	83.0%	90.6%	6
Dataset C	Rolling Z-Score (365 Days)	41.5%	76.4%	88.1%	7

Dataset A and Dataset B produce nearly identical results. However, Dataset B explains slightly more variance within the first four principal components, suggesting that the longer 180-day growth horizon may capture market regimes somewhat more effectively.

Dataset C shows a different behaviour. The rolling Z-score transformation distributes the variance more evenly across the principal components and therefore requires seven

components to exceed the 90% explained variance threshold. This indicates that the Z-score transformation preserves more feature-specific and anomaly-related information, while the growth-based datasets concentrate a larger share of the variance into fewer common factors.

Overall, all three datasets keep meaningful economic structure and provide a suitable basis for the next step: clustering.

5. Clustering Analysis and Results

5.1 Algorithms and Clustering Strategies

The goal of this project is to identify different market regimes within Bitcoin Timeseries using Unsupervised Machine Learning. Unlike supervised learning, no predefined labels such as "Bull Market", "Bear Market" or "Accumulation Phase" are available. Instead, the goal is to identify such potential structures and market regimes by algorithms directly from the data.

To reduce noise and multicollinearity, PCA as a dimensionality reduction technique was applied before. Based on the previous results, 6 principal components were selected for Dataset A and B, while 7 principal components were selected for Dataset C. These transformed datasets will now be used as the input for the clustering analysis, with every Dataset being tested with all chosen clustering algorithms.

The Algorithms

Three different clustering algorithms will be evaluated:

1. K-Means

K-Means divides the observations into a predefined number of clusters by minimizing the distance between observations and their cluster centers. It assumes relatively compact and spherical clusters and therefore serves as a useful baseline method.

2. Gaussian Mixture Models (GMM)

Gaussian Mixture Models assume that the data is generated from a mixture of multiple probability distributions. Unlike K-Means, GMM provides probabilistic cluster assignments, allowing observations to belong partly to multiple clusters. This may be useful if transitions between Bitcoin market regimes are gradual rather than clearly separated - a hypothesis that is very likely given the mostly gradual moves of assets and prices, especially during transitional phases.

3. HDBSCAN

HDBSCAN is a density-based clustering algorithm that identifies regions with high observation density while treating less dense observations as noise. Unlike K-Means and GMM, the number of clusters does not need to be specified in advance.

Initially, I considered DBSCAN as the density-based clustering algorithm. However, after analyzing different algorithms, I decided to select HDBSCAN as it is generally regarded as a more robust version of DBSCAN. DBSCAN requires a fixed neighbourhood radius (epsilon), which can be difficult to choose when clusters have different densities. Since Bitcoin market regimes can vary over time a single density threshold may not be suitable. For example are bull markets often characterized by high volatility and explosive spikes upwards, while bearmarkets typically follow a calm, low volatility phase lacking spikes. A single fixed neighbourhood radius could fit the bullmarket structure well, but fail the bear market structure completely and vice versa.

HDBSCAN addresses this exact problem by analyzing multiple density levels and automatically selecting the most stable clusters for different density levels. There are still parameters that can

be specified, but opposed to DBSCAN no manual choice of the epsilon is necessary, as HDBSCAN automatically optimizes its clusters on possibly local levels. Additionally, in contrast to K-Means and GMM, HDBSCAN automatically determines the number of clusters directly from the data.

Validation Approach and Evaluation

For **K-Means** and **GMM**, the number of clusters will be selected using a combination of:

- **Elbow Method** (only for K-Means): Measures how much variance remains within the clusters. The optimal number of clusters is usually chosen at the point where additional clusters only provide limited improvements.
- **AIC and BIC Information Criteria** (Only for GMM): Measure how well the model fits the data while penalizing additional complexity. Lower values indicate a better clustering model, with BIC penalizing additional complexity with a stronger penalty compared to AIC.
- **Silhouette Score**: Measures how similar observations are to their own cluster compared to other clusters. Higher values indicate better separated and more compact clusters.
- **Davies-Bouldin Index**: Measures the balance between cluster compactness and separation. Lower values indicate better clustering results.

For the **HDBSCAN**, the main parameter that can be specified is the minimum cluster size. I will test the minimum sizes of 50, 100 and 200 since they seem considerable

Afterwards, the resulting clusters will be visualized with time-series plots to check whether the identified regimes correspond to economically meaningful and temporally connected regime periods. After all, this will be the main evaluation criteria as suitable regimes are not defined by thresholds or optimizations but by economic reasoning and domain interpretation.

Expectations and Hypotheses

Bitcoin is characterized by recurring market cycles, periods of rapid growth, timewise enormous speculation and herding behaviour, crashes and longer accumulation phases. Therefore, I expect that the algorithms identify multiple distinct regimes rather than a simple bull-versus-bear classification.

Additionally, I expect the HDBSCAN to perform best as the different regimes should be especially differentiable by different densities per regime, representing exactly what HDBSCAN was built for.

Considering the datasets, I expect Dataset C to perform worst with K-Means and GMM due to its higher volatility with several variables while I presume that HDBSCAN will be able to handle the difference in volatility better. I generally expect Dataset B to deliver the best results as it seemed to possibly better capture regime-like structures than Dataset A.

5.2 K-Means

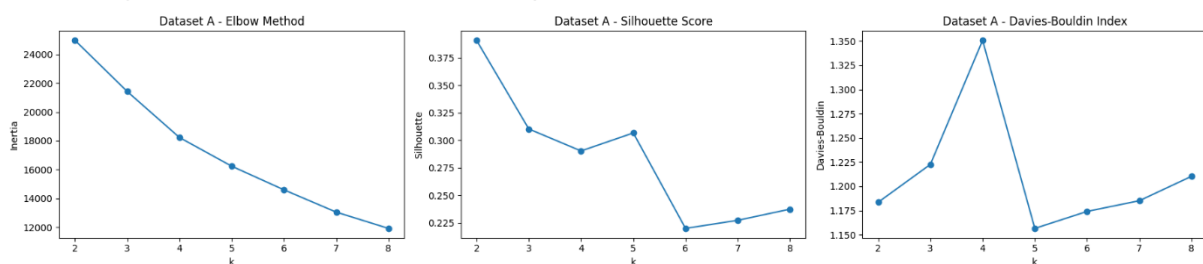
The K-Means process consists of optimizing for the number of clusters first, then applying the algorithms and lastly evaluating the different results per Dataset and per number of clusters.

5.2.1 Optimization for Number of Clusters

For the optimization of the number of clusters for K-Means, I will choose the 2-3 most reasonable ones for each Dataset based on the results of the elbow-method, Silhouette-Score, Davies-Bouldin Index and Economic Reasoning. Afterwards, their clustering results will be evaluated by looking at the visualized clusters compared to the Bitcoin Price Movements.

Number of Clusters for Dataset A

Figure 9: Visualization of Cluster Optimization Criteria for Dataset A



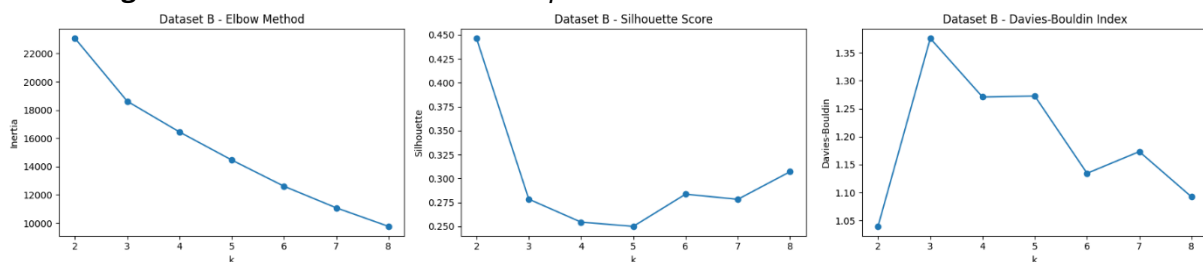
Conclusion:

- Elbow Method: 4 seems to be ideal
- Silhouette-Score: clearly 5
- Davies-Bouldin Index: Clearly 5

→ n(Cluster) of **4 and 5** will be taken as best choices.

Number of Clusters for Dataset B

Figure 10: Visualization of Cluster Optimization Criteria for Dataset B



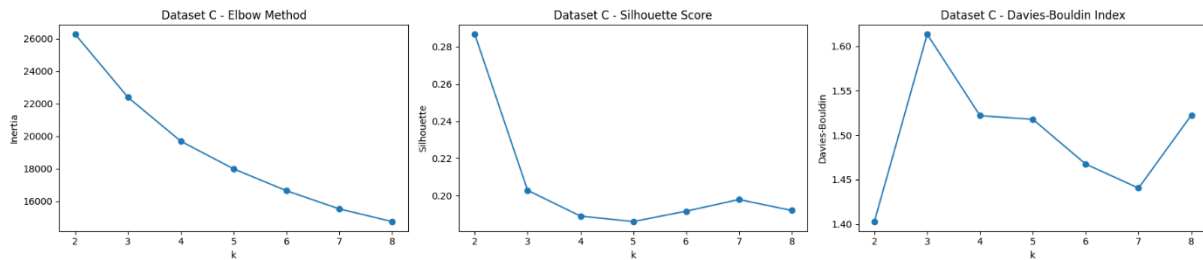
Conclusion:

- Elbow Method: 3 seems to be ideal
- Silhouette-Score: clearly 8
- Davies-Bouldin Index: Clearly 8

→ n(Cluster) of **3 and 8** will be taken as best choices.

Number of Clusters for Dataset C

Figure 11: Visualization of Cluster Optimization Criteria for Dataset C



Conclusion:

- Elbow Method: 4 seems to be ideal
- Silhouette-Score: clearly 7
- Davies-Bouldin Index: Clearly 7

→ n(Cluster) of **4 and 7** will be taken as best choices.

Accordingly, the following number of Clusters in the corresponding Datasets will be further evaluated:

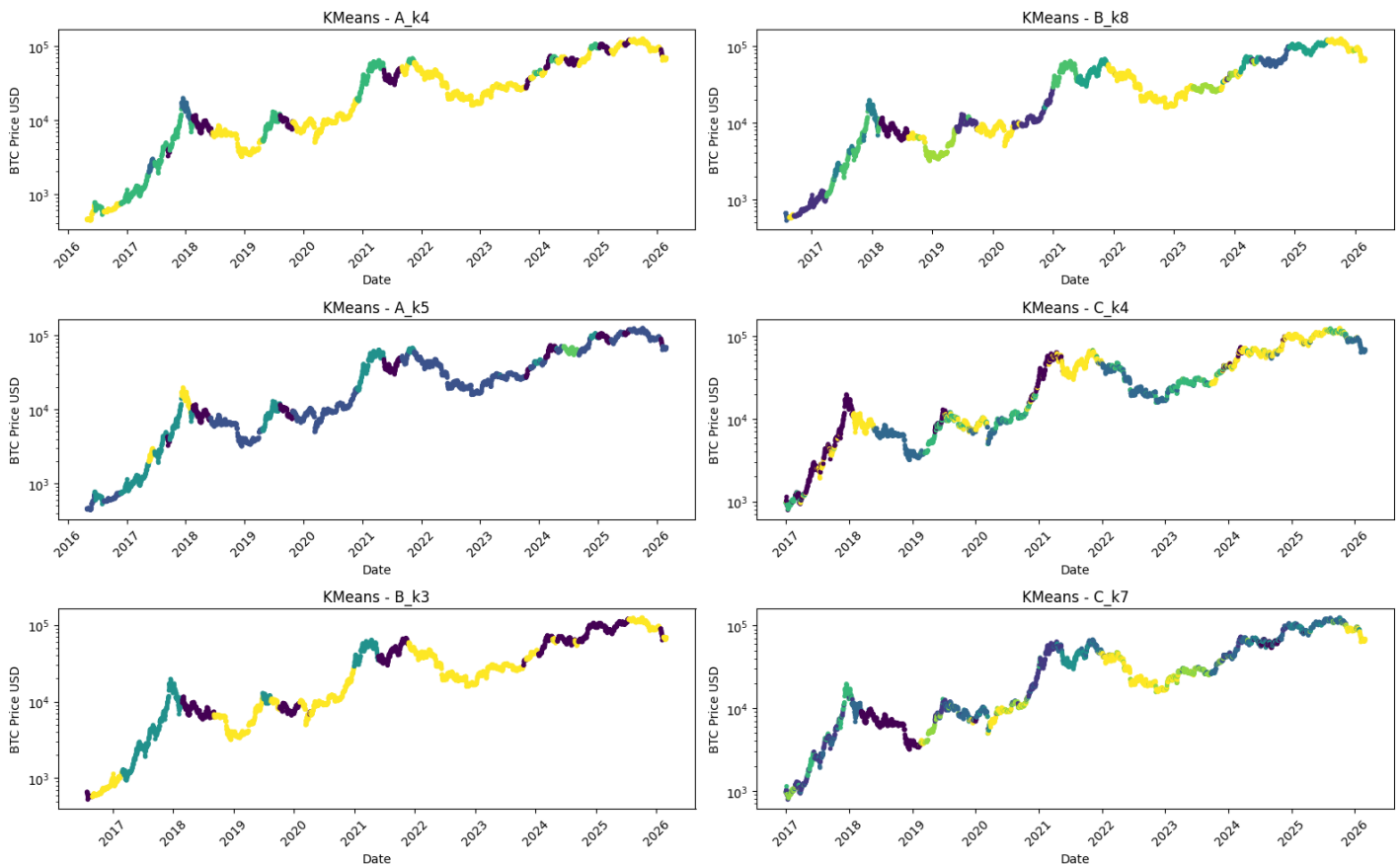
- **A:** 4, 5
- **B:** 3, 8
- **C:** 4, 7

5.2.2 Cluster Results and Evaluation

After first plotting the 1st Principal Components against the 2nd and visualizing the pure clusters over time (see Appendix B), I had a look at the main evaluation criteria: How well the clusters actually capture the probable regimes in Bitcoin over time.

The results are generally incredible and exceed any expectation! I've seen people trying to plot the different market regimes and market phases hundreds of times but I'm not sure if I've ever seen plots also incorporating signals for upcoming changes that early on. Generally, all plots showcase very clear regimes of bear and bullmarkets, the transitional phases in between, and depending on the number of clusters additional peak phases as well. K-Means generally found temporally connected phases without the usage of explicit time encoding.

Figure 12: BTC Price over Time coloured by K-Means Clusters , per Dataset and chosen Clusters



Dataset A Results:

While consistently capturing Bull Phases and its peaks, the clusters of A show with both 4 and 5 Clusters a strong imbalance favouring bear market phases and generally unclear regimes in the latest data. Possibilities of underfitting appear as well in the share of observations per cluster.

Dataset B Results:

Dataset B generally seems to capture important market transitions better, having less imbalance towards bear markets, while at the same time, for both 3 and 8 Clusters showing same clusters in different periods of time that actually do not look like the same market regime. The verdict is: better than A, but not fully convincing yet. Longer time horizons for Growth rates seem to indeed capture the regimes better as was expected.

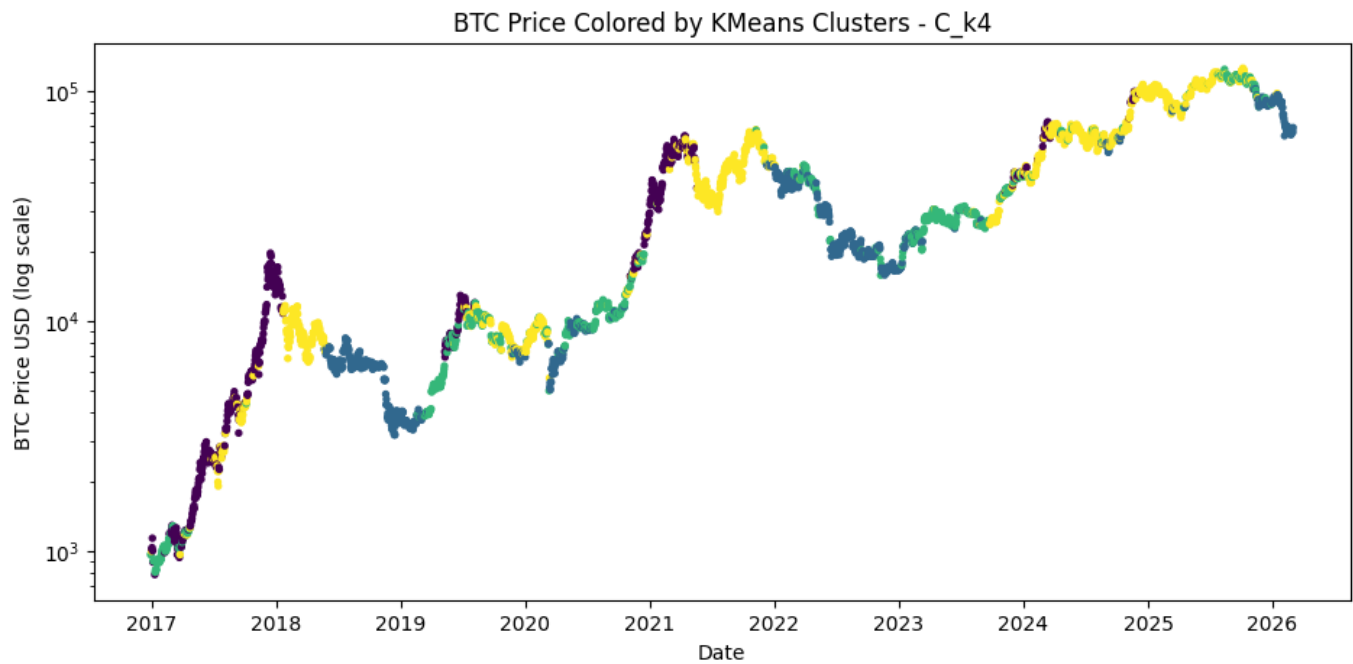
Dataset C Results:

The Results of this Dataset are by far the best. Not only are for both 4 and 7 clusters the selection for regimes across different periods of time consistent, but also are the upcoming transitional periods and changes from bull to bear markets and vice versa noticeable very early on, especially with 4 clusters. In example would this method had given an early warning signal of changing clusters during peaks of every bull run, sometimes even up to 3 months ahead of a sell-off. Comparing both 4 and 7 clusters, 4 performs better as 7 clusters seem to overfit slightly, while also being less reliable signaling upcoming regime changes.

5.2.3 Conclusion K-Means

By applying the K-Means algorithm, Dataset C results in the most useful clusters. While 7 clusters seem to slightly overfit, 4 clusters balances the different regimes while also indicating future regime changes very well. Generally, the clusters are very coherent, temporally connected and can even act as early warning signals.

Figure 13: Best performing K-Means Model (K-Means C-k4)



5.3 Gaussian Mixture Models

The process for applying and evaluating GMM is mostly analogous to K-Means: I will first optimize for the number of clusters, then apply the algorithm and lastly evaluating the different results per Dataset and Number of Clusters.

Two things are different:

1. The AIC/BIC typically replaces the elbow-method in optimizing for the number of clusters as the GMM is a probability-driven approach.
2. The results do not just show the belonging of datapoints to clusters, but the probability of it too, while also giving probabilities for belonging to other clusters. This is especially interesting as it could reveal transitional phases or early signals for them as a higher uncertainty and with that, gives a possible additional useful dimension of the clusters.

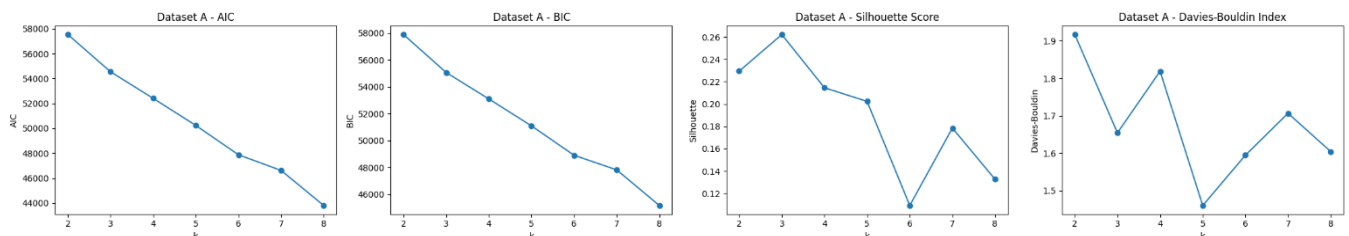
5.3.1 Optimization for Number of Clusters

I will again choose for the optimization of the number of clusters for GMM analogous the 2-3 most reasonable ones for each Dataset based on the results of the elbow-method, Silhouette-Score, Davies-Bouldin Index, Economic Reasoning, and *usually* AIC/BIC scores.

I've seen that generally with any Dataset, the AIC/BIC gets lower the more features are added. Based on experience on other projects with AIC/BIC, I know that the metric can be very misleading and will therefore not focus on it - if any, I will take a possible *elbow* inside these functions as a hint as to where adding more features gets less efficient.

Number of Clusters for Dataset A

Figure 14: Visualization of Cluster Optimization Criteria for Dataset A



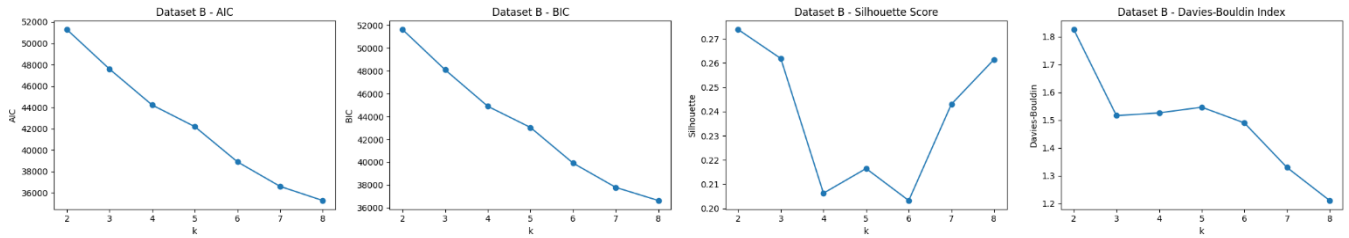
Conclusion:

- AIC/BIC - elbow: 3 or 6
- Silhouette-Score - the higher the better: Clearly 3
- Davies-Bouldin Index - the lower the better: Clearly 5

→ n(Cluster) of **3 and 5** will be taken as best choices.

Number of Clusters for Dataset B

Figure 15: Visualization of Cluster Optimization Criteria for Dataset B



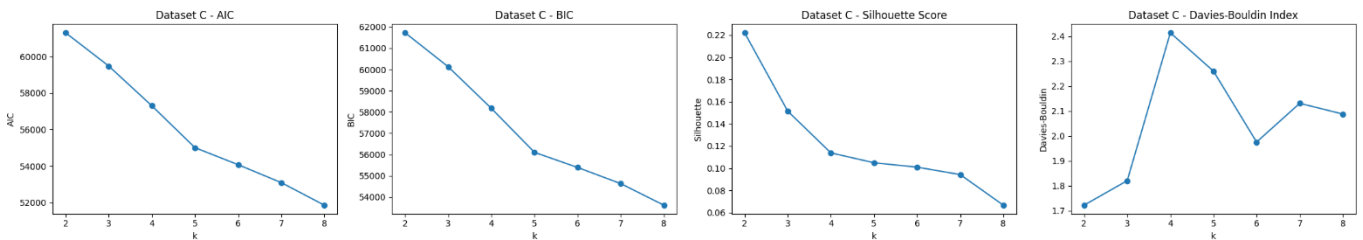
Conclusion:

- AIC/BIC - elbow: 4
- Silhouette-Score - the higher the better: 3 or 8
- Davies-Bouldin Index - the lower the better: 3 or 8

→ n(Cluster) of **3 and 8** will be taken as best choices.

Number of Clusters for Dataset C

Figure 16: Visualization of Cluster Optimization Criteria for Dataset C



Conclusion:

- AIC/BIC - elbow: 5
- Silhouette-Score - the higher the better: 3 or 4
- Davies-Bouldin Index - the lower the better: 4 or 5

→ n(Cluster) of ****4 and 5**** will be taken as best choices.

Accordingly, the following number of Clusters in the corresponding Datasets will be further evaluated:

- **A:** 3, 5
- **B:** 3, 8
- **C:** 4, 5

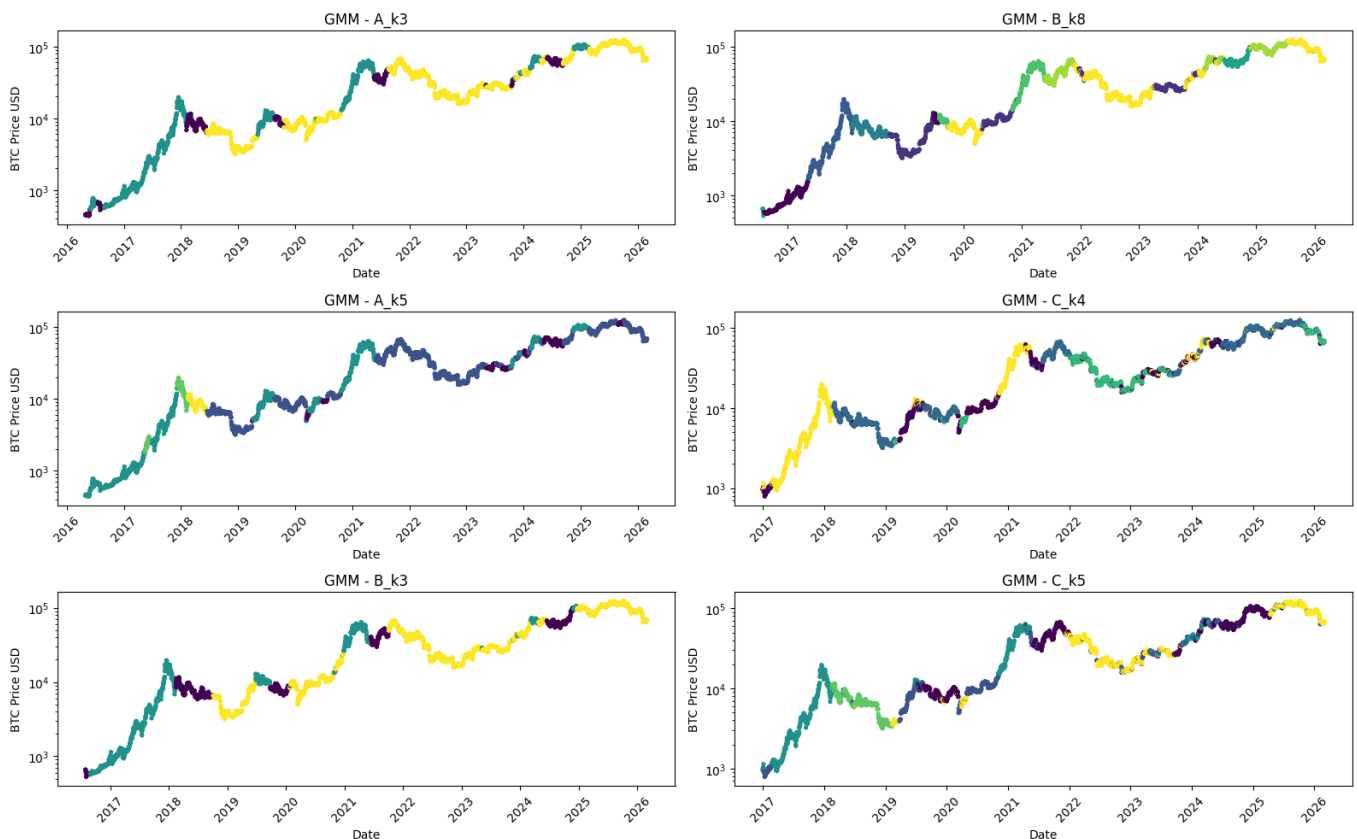
5.3.2 Cluster Results and Evaluation

Cluster-based Results

The results are generally very similar to the ones with K-Means, also considering the different Datasets. All plots showcase very clear regimes of bear and bullmarkets, the transitional phases in between, and depending on the number of clusters additional peak phases as well. GMM too generally found temporally connected phases without the usage of explicit time encoding.

Compared to K-Means, the main difference that stands out is that GMM does not find signals for upcoming market regime changes as consistent and as early as K-Means, making K-Means the preferred algorithm for now. I will discuss that in the discussion and comparison about the different algorithms.

Figure 17: BTC Price over Time coloured by Clusters from GMM, per Dataset and chosen Clust.



In Detail:

Dataset A Results:

GMM consistently captures the different regimes, without the imbalance favouring bear market phases as visible in K-Means. While 5 clusters seem to slightly overfit, both seem to develop inconsistent clusters within the latest period with several changes over similar regimes.

Dataset A Results:

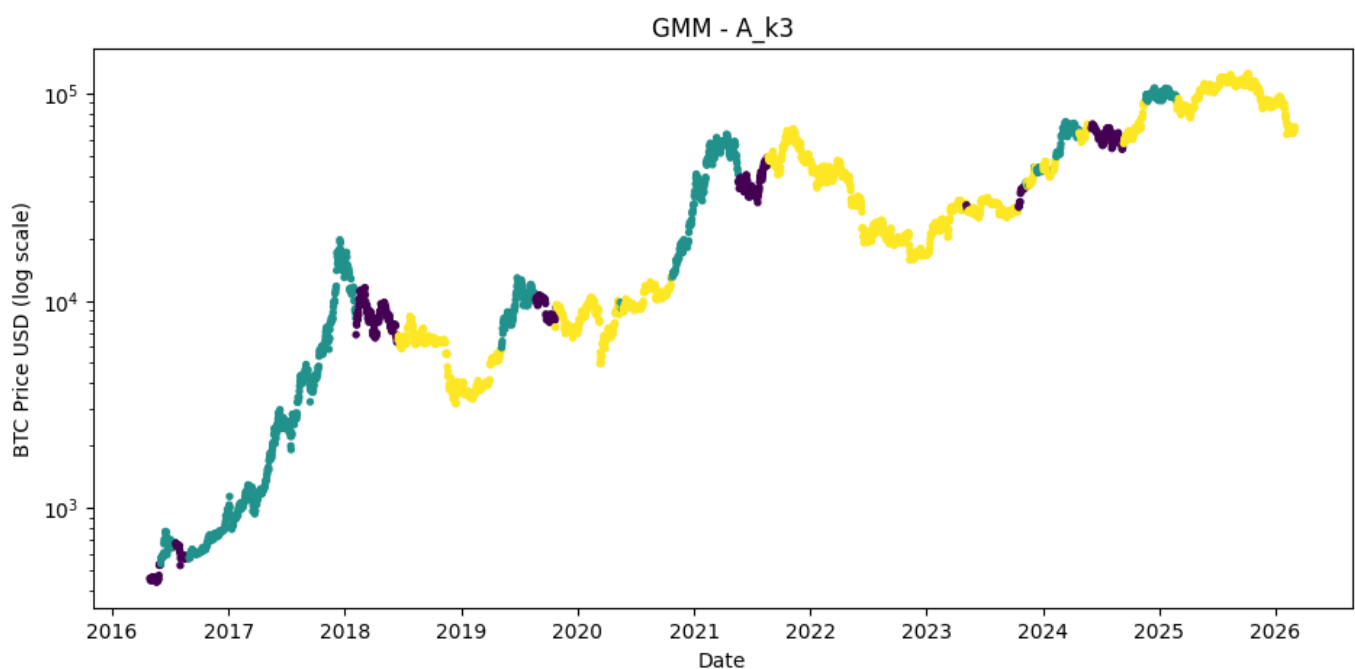
Dataset B generally seems to capture important market regimes and transitions worse. Both 3 and 8 clusters appear to cluster more incoherently the newer the data, with 8 clusters seeming to overfit. Dataset A with 3 clusters stays the best approach.

Dataset C Results:

While looking similar to the results with K-Means at first glance, the results here are the most unreliable. F.e. pre- and post-Bullrun Peaks are clustered together, not representing the transitions to completely different market regimes.

The preferred model within GMM would be based on Dataset A with 3 clusters, while at the same time clearly outperformed by the prior best K-Means Model.

Figure 18: Best performing GMM Model (GMM A-k3)



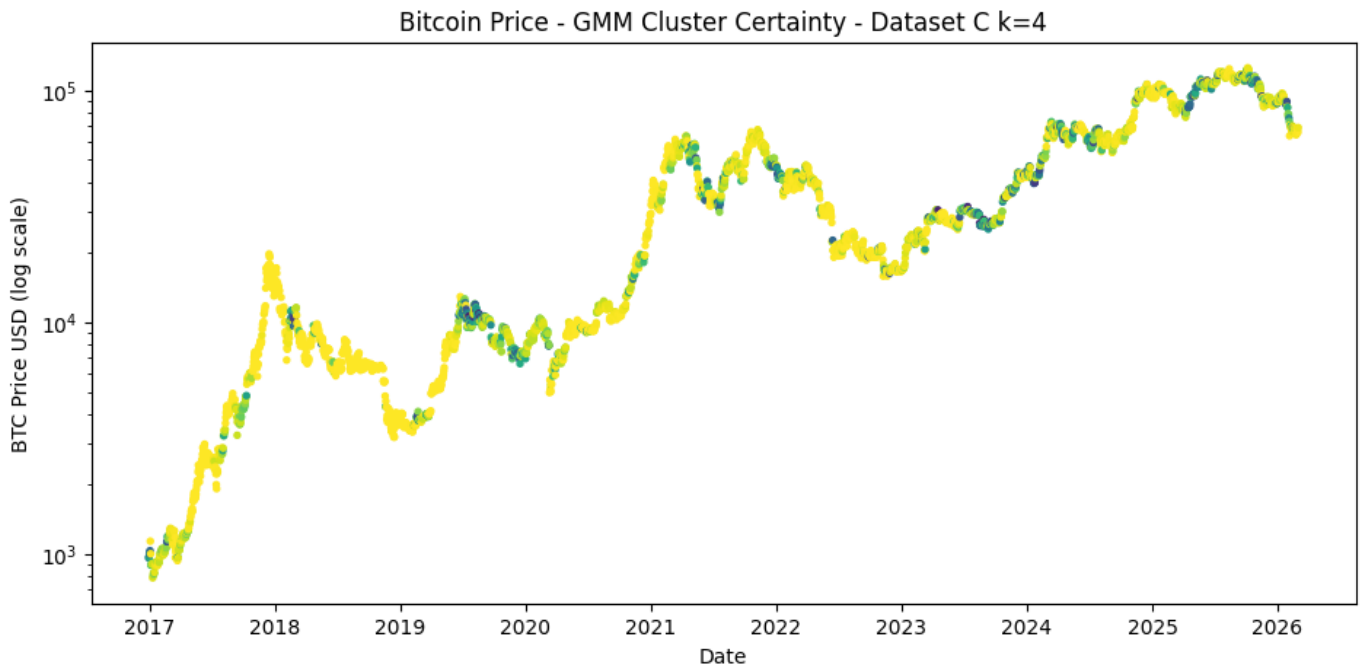
Confidence-based Results

GMM allows additionally to have a glimpse into the confidence of the algorithm for each datapoint to be in the given cluster. I can use this to possibly find signals indicating upcoming transitions from one regime to another, as these phases are usually driven by high uncertainty in the real world as well. This stays exploratory, as the main focus is on the clusters for different regimes as such.

Importantly, I did not use the prior best model here, as I'm looking for early regime changes and transition signals instead of clear regimes. I'm looking for uncertainty in the data with unclear borders between regimes because exactly this uncertainty will be visualized. I tried the A-k3 model prior here and it mostly showed a very high confidence, with some small exceptions that were usually located in the midst of the actual transition.

I therefore tested further and found C-k4 to be the best model to find early signals for upcoming transition periods by its uncertainty and focus the following discussion on the results of this model here.

Figure 19: BTC Price over time coloured by GMM-certainty (GMM C-k4)



These results are very interesting. Not only do we see clear differences between clear bull and bearmarkets and the transitional phases in between within the GMM C-k4, but also early signals, sometimes before the prices started swinging into the other direction. While exploratory, combining these clusters as an additional signal for regime changes with the preferred general cluster model of K-Means could enhance the overall value.

5.3.3 Conclusion GMM

The GMM Model A-k3 performs the best, while at the same time still suffering from inconsistency of clustering market regimes across different periods of time, especially in the latest period. K-Means C-k4 is still preferred by far.

Additionally, the Uncertainty Clusters of GMM C-k4 reveal possible additional information considering early signals of upcoming transitions. Interestingly, the same Dataset and Cluster Size as the preferred K-Means Model, indicating higher information value in terms of important signals.

5.4 HDBSCAN

The HDBSCAN process differs to the prior two in that it will find the ideal number of clusters automatically. The only Parameter that can be tweaked is the minimum size of clusters. I applied 25, 50 and 100 for the minimum size of clusters in a first round, but had unsatisfactory results: 100 would often only lead to a single cluster, 25 seemed to split everywhere, while 50 went seemed to underfit the data. I therefore tried again with 30, 35, 40, 45 as minimum size of clusters. The results for optimization-metrics such as Silhouette-Scores and the Percentage of Noise left got much better, but the visualized timeseries only worse. I therefore show this chapter as applied with 25, 50 and 100 for the minimum size of clusters.

5.4.1 Optimization for Minimum Cluster Size

Table 4: Criteria for ideal minimum cluster size

<i>Dataset</i>	<i>Dataset Min Cluster Size</i>	<i>Number of Clusters</i>	<i>Noise Share</i>	<i>Silhouette Score</i>	<i>Davies-Bouldin Index</i>
<i>Dataset A</i>	25	18	0.203	0.247	0.680
<i>Dataset A</i>	50	4	0.566	0.369	0.654
<i>Dataset A</i>	100	0	1.000	-	-
<i>Dataset B</i>	25	22	0.318	0.481	0.736
<i>Dataset B</i>	50	7	0.199	0.318	0.765
<i>Dataset B</i>	100	2	0.269	0.281	0.905
<i>Dataset C</i>	25	2	0.816	0.375	1.157
<i>Dataset C</i>	50	2	0.780	0.363	1.110
<i>Dataset C</i>	100	2	0.923	0.565	0.726

Given that the share of noise should be minimized and the number of cluster in a reasonable range for clustering market regimes (3-8), the following can be concluded:

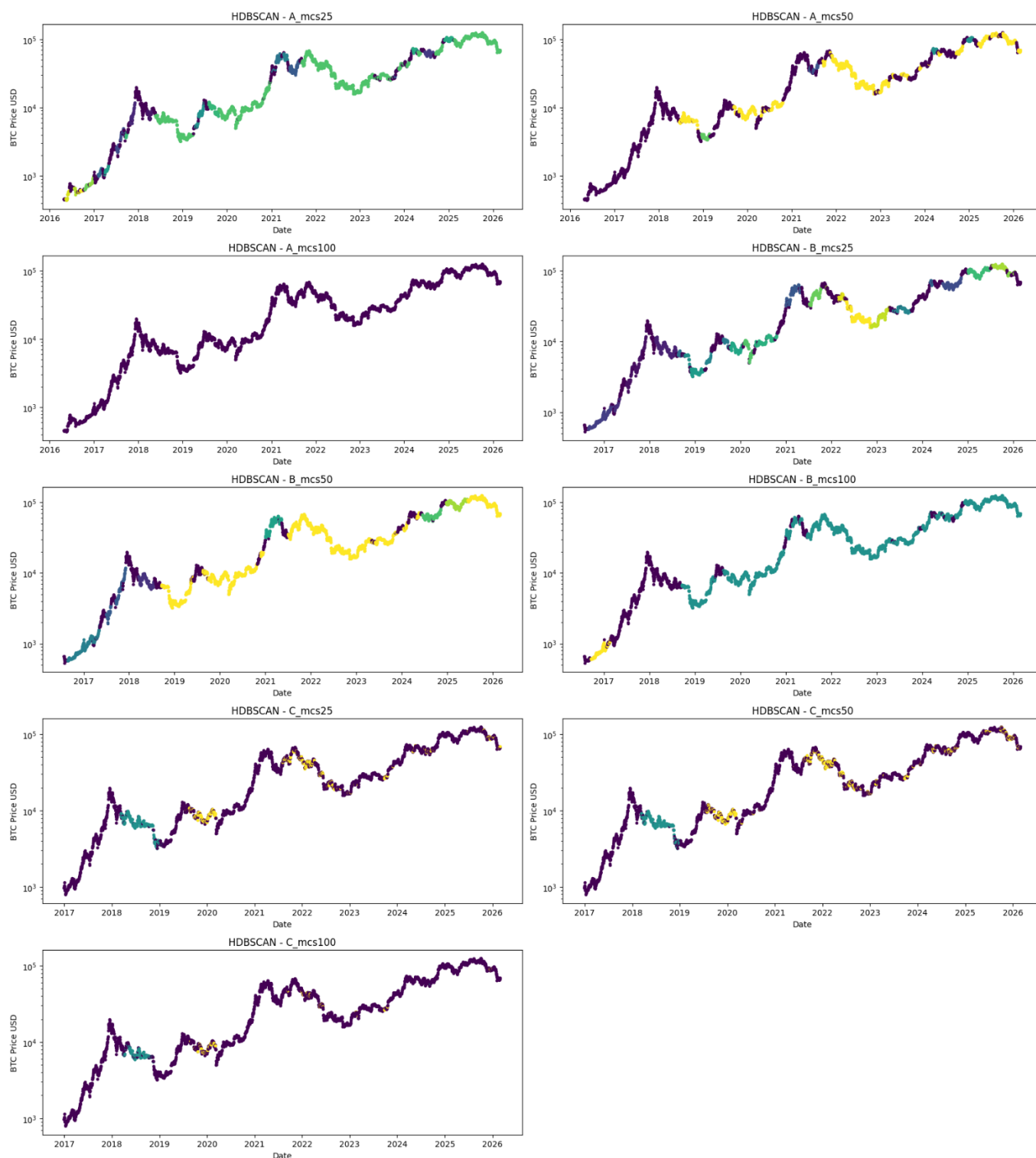
- The only Models with a considerable noise share would be A-25 (20%), B-50 (19%) and maybe B-100 (26%).
- While A-25 overfits with 18 clusters and B-100 underfits with only 2, it is expected that B-50 shows the best results.

Note for the 2nd try with 30, 35, 40, 45: A-35 was very interesting; 3 clusters, only 4% noise share. B-45 had 6 clusters and only 16% noise. While the rest clearly either under or overfitted, in the end all visualizations failed to capture regimes as accurate as the prior algorithms.

5.4.2 Cluster Results and Evaluation

Overall, the results were considerably less convincing than the previous clustering approaches. Across all three datasets and parameter settings, HDBSCAN either produced only one dominant cluster or isolated a few small periods while assigning the majority of observations repeatedly to a single regime, while always suffering from inconsistently clustering different regimes together over the whole time period. To some extent, some specifications do find bull and bear market regimes, but over- and underfitting combined with incoherency make the results not useful to confidently identify market regimes and possible transitional periods.

Figure 20: BTC Price over Time coloured by Clusters from HDBSCAN, per Dataset and Min Cl. S.



Every result was inconsistently labelling similar market regimes with different clusters over time.
The additional details excluding coherence:

Dataset A Results:

A general tendency of underfitting, with A-25 focusing on bear markets, to A-100 only showing a single cluster.

Dataset B Results:

A good fitting of B-25 while showing very strong inconsistency, while developing more and more underfitting the higher the minimal clustering size.

Dataset C Results:

All specifications clearly underfit. Rolling Z-Scores seem to have produced very similar Density-results, what is reasonable given that it directly incorporates statistical normal distributions compared to growth rates.

5.4.3 Conclusion HDBSCAN

The HDBSCAN results suggest that density-based clustering is not well suited for the selected feature representations. In contrast to K-Means and, to a lesser extent, GMM, HDBSCAN was generally unable to identify consistent and interpretable Bitcoin market regimes.

Increasing the minimum cluster size further reduced the number of detected clusters, while smaller values mainly resulted in a few isolated clusters without improving overall interpretability.

No model would be preferred here as no model reliably identifies regimes. K-Means C-k4 is therefore considered the best model.

6. Discussion and Comparison of Algorithms

6.1 General Expectations and Findings

The objective of this project was to identify meaningful Bitcoin market regimes using Un-supervised Machine Learning techniques. Overall, the results strongly support the hypothesis that Bitcoin market behaviour consists of multiple repeating market regimes rather than a simple bull-versus-bear classification. Across most successful specifications, the algorithms identified not only bull and bear markets, but also transitional phases in between possibly representing accumulation and distribution phases and, in some cases, separate peak or euphoria phases.

One of the most important findings of this project is that feature engineering had a large impact on the final results, to some extent even larger than the clustering algorithm itself - with some clear exceptions. While all clustering methods were applied to the same underlying information, the choice of feature representation significantly changed the resulting regime structures.

The initial expectation was that Dataset B would perform best because the longer growth horizon appeared to capture regime-like structures more effectively during the exploratory analysis. Surprisingly, Dataset C consistently produced the strongest results. The rolling Z-Score transformation seems to capture the different market regimes across time more effectively than Moving-Average-enhanced growth rates. Instead of measuring how much a variable changed, it measures how unusual the current observation is relative to its historical behaviour, that is 365 days for this case. This seems particularly useful for identifying regime changes and market transitions.

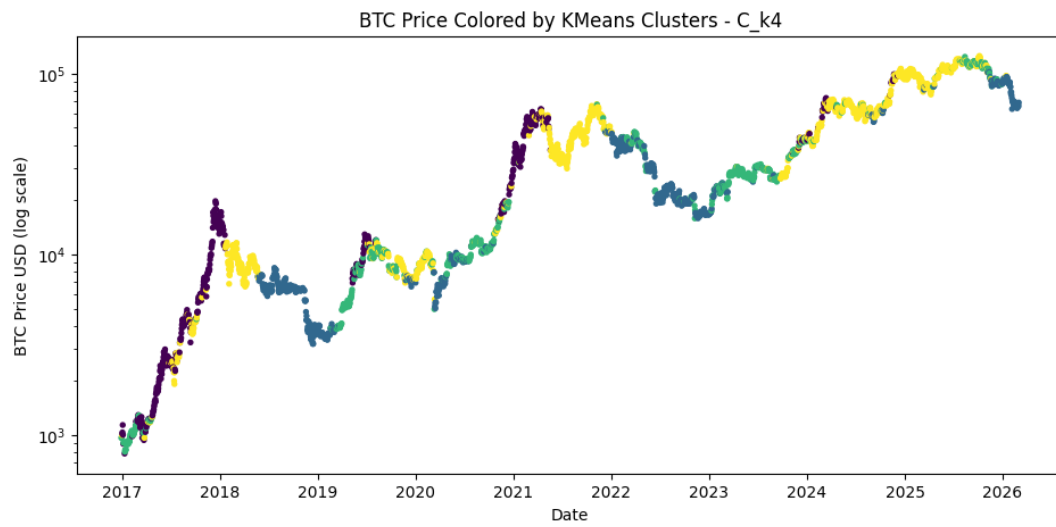
6.2 Comparing Algorithms

Comparing the clustering algorithms (see Figure 21), K-Means produced the most convincing and interpretable results. The preferred specification, Dataset C with 4 clusters, generated consistent and temporally connected market regimes while also providing surprisingly early signals of upcoming market transitions. In multiple cases, cluster changes were signaled weeks or even months before major market reversals became visible in price action. While these observations remain exploratory, they suggest that the selected feature representation contains valuable information about changing market conditions.

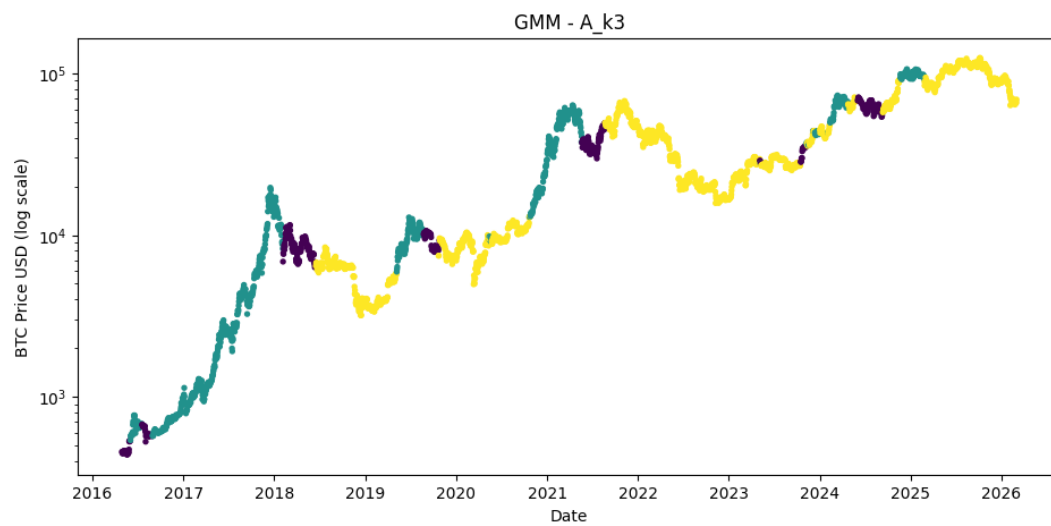
Gaussian Mixture Models produced results that were generally similar to K-Means. The identified market regimes were often comparable and the overall structure remained visible across datasets. However, GMM struggled more with assigning consistent regimes across different periods of time. Similar market environments were occasionally assigned to different clusters, while clearly different market conditions sometimes appeared within the same cluster. As a result, the economic interpretation of the clusters was generally weaker than under K-Means.

Figure 21: Comparing the best models across K-Means, GMM and HDBSCAN

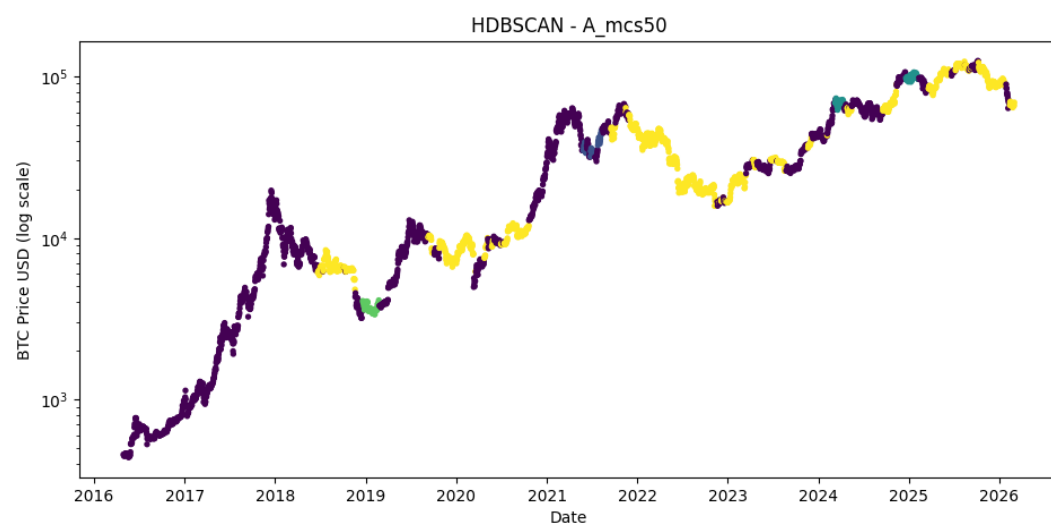
K-Means



Gaussian Mixture Model



HDBSCAN



Nevertheless, GMM provided another interesting additional insight: The possibility of visualizing GMM's probabilities of datapoints belonging to different clusters can be used to detect phases of higher uncertainty, which typically often during important transition phases and turning points in markets. While the clustering performance itself was worse than K-Means, the uncertainty estimates could contain useful information regarding potential future regime transitions. Combining the regime classification of K-Means with the uncertainty information from GMM could therefore be an interesting area for future research.

The largest surprise of the project was the performance of HDBSCAN. Based on the theoretical properties of Bitcoin market cycles, the expectation was that density-based clustering would perform particularly well. Different market regimes were expected to form distinct density regions within the feature space, making HDBSCAN a suitable candidate. However, the results showed the opposite. Across all datasets and parameter settings, HDBSCAN struggled to identify stable and interpretable regime structures. Most specifications either collapsed into one dominant cluster or isolated a few small periods without generating a consistent overall regime classification.

This suggests that Bitcoin market regimes may not form clearly separated density clusters within the selected feature space. Instead, the market appears move more gradually through different regimes in the selected feature space, making centroid-based methods such as K-Means more suitable than density-based approaches.

Overall, the results indicate that meaningful Bitcoin market regimes can be identified reliably using Unsupervised Machine Learning. But the success of the approach depends heavily on the feature engineering process and the representation of the underlying information. Among all tested combinations, K-Means on Dataset C with 4 clusters (C-k4) provided the most convincing results.

The previous results suggest that K-Means on Dataset C with 4 clusters provides the most meaningful representation of Bitcoin market regimes. However, the clustering results alone do not indicate whether these regimes also form visible structures within the underlying feature space. To investigate this further, a t-SNE visualization was created for the preferred model. As a nonlinear dimensionality reduction technique, t-SNE allows a visual inspection of whether observations belonging to the same cluster also tend to group together in a lower-dimensional representation.

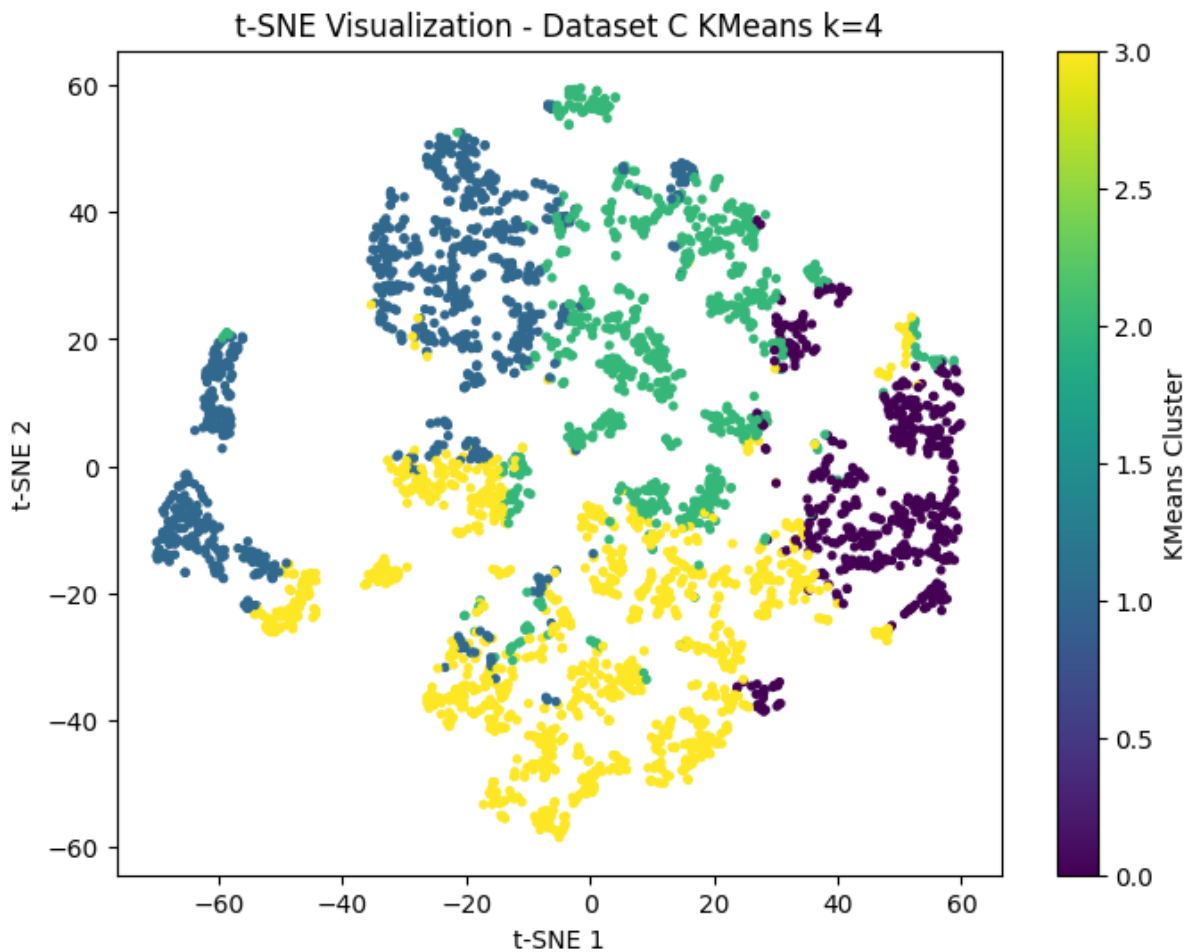
6.3 t-SNE Visualization

While PCA was used for dimensionality reduction before clustering, PCA is a linear method and may therefore miss nonlinear structures within the data. To investigate whether additional nonlinear structure exists and therefore, could be used to enhance future research, t-SNE was applied as an exploratory visualization technique on the best performing model - K-Means C-k4.

Unlike PCA, t-SNE focuses on preserving local neighbourhood relationships and is often used to visualize high-dimensional datasets in two dimensions. It was not used for clustering itself, but rather as an additional validation tool to check whether the selected clusters also appear as meaningful groups in a nonlinear representation of the feature space.

The resulting plot shows that the identified clusters generally form distinct regions within the embedded feature space. While the clusters are not perfectly separated, clear grouping structures can still be observed. This is a positive result, as Bitcoin market regimes are not expected to represent completely isolated states but rather gradual transitions between different market environments.

Figure 22: *t-SNE Visualization of the best performing Model (K-Means C-k4)*



Some overlap between clusters remains visible, particularly around cluster boundaries. This is consistent with the previous findings that market transitions often occur gradually and that observations close to turning points may share characteristics of multiple regimes at the same time.

Overall, the t-SNE visualization provides additional support for the selected K-Means solution. The identified clusters appear to capture meaningful underlying structure in the data rather than representing purely random partitions of the feature space. While at the same time, these results indicate that nonlinear structure does exist, and non-linear dimensionality reduction techniques such as UMAP (Uniform Manifold Approximation and Projection) could be used to possibly enhance the results even further.

6.4 Future Research and Possible Improvements

Several promising extensions and possible improvements could not be explored within the scope of this project due to time constraints, but can possibly be explored at later stages.

The most important limitation of the current approach is the available feature set. While the dataset already combines market, network and on-chain variables, Bitcoin offers substantially more information than what was included, as on-chain data can track any movements and deliver for example signals of whale-holder movements (Bitcoin-holders with very high amounts of BTC) that typically come with higher impacts on market moves. Future work could incorporate additional on-chain metrics such as whale holdings and movements, exchange transfer activity, wallet distribution measures, realized capitalization metrics and other indicators that may contain useful information about changing market conditions. At the same time, macroeconomic factors such as monetary policy and global stock developments could carry additional information and could further enhance the current results of this project.

Another very important challenge, and with it the possible the easiest-to-achieve improvement with highest impact, is the long-term trend of diminishing returns in Bitcoin. Early bull markets were significantly stronger than more recent cycles, resulting in market peaks that keep getting lower and lower, although representing similar market regimes. This may partially explain why some clusters appear less consistent across different periods. Future work could attempt to explicitly model this phenomenon and normalize the data accordingly before feature engineering and clustering by approximating the long-term Bitcoin function and re-scale the data accordingly.

Further, the rolling Z-Score transformation produced the strongest overall results, but several variations remain unexplored. Different rolling windows could be tested instead of the current 365-day specification. Additionally, the Z-Score could be calculated relative to a moving average rather than the current observation itself. This may reduce short-term noise and potentially improve the identification of transitional market phases.

The current analysis also relies on PCA dimensionality reduction. Future work could investigate whether clustering directly on the engineered feature space could lead better results, as the PCA-reduced dataset featured 6 features compared to the prior only 10 or 11 (for Dataset C). Furthermore, alternative dimensionality reduction approaches could be explored, including fewer principal components or nonlinear techniques such as UMAP to capture the transition periods better.

Several variables were retained despite some uncertainty regarding their contribution to the final model. Features such as issuance and total transaction fees could be removed individually and the resulting changes in clustering performance could be evaluated systematically. From a methodological perspective, additional clustering approaches may also be promising. In particular, Hidden Markov Models appear well suited for regime detection problems and could potentially model market transitions more naturally than static clustering algorithms.

Finally, the identified market regimes could be used as a foundation for more advanced predictive systems. One possible approach would be to train autoencoders on specific market regimes, such as bull and bear markets, and use reconstruction errors as early warning indicators for regime transitions. This could potentially improve the ability to detect changing market conditions

before they become visible in price action and act as a reliable signal to spot changes in market trends before they actually appear.

7. Conclusion

This project investigated whether Unsupervised Machine Learning techniques can identify meaningful Bitcoin market regimes using market, network and on-chain data. Three different feature engineering approaches were developed and compared, consisting of

- a. **90 day growth rates** based on 30 day moving averages,
- b. the same approach with **180 day growth rates** instead, and
- c. **Rolling Z-Scores** based on 365 day windows

for the variables that necessitated transformation. The results demonstrate that feature engineering is a critical component of the clustering process and might have, to some extent, a larger influence on the final outcome than the clustering algorithm itself. Among the tested approaches, the rolling Z-Score transformation consistently produced the most informative and interpretable feature representation, indicating that Bitcoin Market Regimes are not identified by how much things changed in the past, but in relation to the past.

Principal Component Analysis was used to reduce dimensionality while preserving more than 90% of the original variance, allowing efficient clustering without substantial information loss. Based on the newly created PCA features, three clustering algorithms were evaluated:

1. **K-Means** produced the strongest overall results and generated consistent, temporally connected market regimes that aligned well with historical Bitcoin market phases.
2. **Gaussian Mixture Models** identified similar structures but generally provided less consistent regime classifications, while possibly allowing for finding possible trend reversal structures in the magnitude of uncertainty of datapoints belonging to specific clusters.
3. **HDBSCAN** failed to identify stable and interpretable regimes and was therefore not considered suitable for the selected feature representation.

The final preferred model consists of K-Means clustering with 4 clusters applied to Dataset C (K-Means C-k4) after PCA dimensionality reduction. This specification produced the most economically meaningful regime structure and showed promising ability to identify regime transitions up to weeks or months before major market reversals occurred.

While the results are promising, several extensions remain unexplored. In particular, richer on-chain feature sets, alternative dimensionality reduction techniques, Hidden Markov Models and approaches that explicitly account for Bitcoin's diminishing-return dynamics may further improve regime identification. These directions provide interesting opportunities for future research. Furthermore, the predictive value of the identified regime changes could be evaluated in a systematic trading or risk management framework. At the same time, several limitations remain: The interpretation of market regimes is inherently subjective and there is no objectively correct number of clusters. Furthermore, the analysis focuses exclusively on Bitcoin-specific variables and does not incorporate macroeconomic, sentiment or cross-asset information.

Overall, the results suggest that unsupervised machine learning provides a useful framework for understanding Bitcoin market behaviour and identifying repeating market regimes. More importantly, the project demonstrates that well-defined and executed feature engineering can uncover meaningful market structures that remain visible across different clustering approaches.

Appendix A: Dimensionality Reduction

Table 5: PCA Loadings Dataset A

Feature	PC1	PC2	PC3	PC4	PC5	PC6
<i>active_addresses_ma30_g90d</i>	0.326	0.276	-0.285	0.095	-0.271	-0.188
<i>nonzero_addresses_ma30_g90d</i>	0.327	-0.006	0.061	0.306	-0.340	0.717
<i>transaction_count_ma30_g90d</i>	0.139	0.564	0.421	-0.017	0.355	0.050
<i>transfer_count_ma30_g90d</i>	0.272	0.554	0.089	0.019	0.129	-0.108
<i>hash_rate_ma30_g90d</i>	0.179	-0.302	0.130	0.781	0.319	-0.339
<i>exchange_supply_btc_ma30_g90d</i>	0.030	-0.253	0.823	-0.207	-0.247	-0.122
<i>exchange_outflow_usd_ma30_g90d</i>	0.383	-0.101	-0.092	-0.262	-0.194	-0.287
<i>price_usd_ma30_g90d</i>	0.393	-0.142	-0.071	-0.146	0.236	-0.087
<i>spot_volume_usd_ma30_g90d</i>	0.394	-0.090	-0.010	-0.102	-0.282	-0.278
<i>mrvr_ratio</i>	0.280	-0.307	-0.101	-0.366	0.577	0.271
<i>total_fees_btc</i>	0.354	-0.111	0.107	0.095	-0.008	0.258

Table 6: PCA Loadings Dataset B

Feature	PC1	PC2	PC3	PC4	PC5	PC6
<i>active_addresses_ma30_g180d</i>	0.298	0.357	-0.268	0.205	-0.021	-0.321
<i>nonzero_addresses_ma30_g180d</i>	0.346	0.053	0.048	0.351	0.787	-0.166
<i>transaction_count_ma30_g180d</i>	0.043	0.528	0.480	-0.192	0.003	0.119
<i>transfer_count_ma30_g180d</i>	0.202	0.575	0.136	-0.057	-0.175	-0.020
<i>hash_rate_ma30_g180d</i>	0.195	-0.276	0.295	0.666	-0.334	0.144
<i>exchange_supply_btc_ma30_g180d</i>	0.039	-0.310	0.717	-0.267	0.163	-0.278
<i>exchange_outflow_usd_ma30_g180d</i>	0.390	-0.132	-0.080	-0.275	-0.282	-0.263
<i>price_usd_ma30_g180d</i>	0.403	-0.128	-0.042	-0.142	-0.136	-0.089
<i>spot_volume_usd_ma30_g180d</i>	0.397	-0.107	0.019	-0.003	-0.210	-0.245
<i>mrvr_ratio</i>	0.314	-0.204	-0.234	-0.428	0.262	0.335
<i>total_fees_btc</i>	0.372	0.031	0.116	0.047	-0.017	0.713

Table 7: PCA Loadings Dataset C

Feature	PC1	PC2	PC3	PC4	PC5	PC6	PC7
<i>active_addresses_z365d</i>	0.305	0.282	-0.154	-0.051	-0.359	-0.007	0.225
<i>nonzero_addresses_z365d</i>	0.224	0.192	-0.325	0.244	0.638	0.284	0.482
<i>transaction_count_z365d</i>	0.163	0.519	0.349	-0.167	0.191	-0.257	-0.178
<i>transfer_count_z365d</i>	0.260	0.507	0.142	-0.151	0.111	-0.154	-0.104
<i>hash_rate_z365d</i>	0.198	0.090	0.454	0.584	-0.426	0.217	0.218
<i>issuance_usd_z365d</i>	0.366	-0.210	0.190	0.274	0.140	-0.157	0.041
<i>exchange_supply_btc_z365d</i>	0.005	-0.329	0.626	-0.437	0.225	0.056	0.433
<i>exchange_outflow_usd_z365d</i>	0.349	-0.107	-0.206	-0.300	-0.206	-0.175	0.152
<i>price_usd_z365d</i>	0.373	-0.249	0.003	0.170	0.119	-0.271	-0.246
<i>spot_volume_usd_z365d</i>	0.337	-0.085	-0.206	-0.329	-0.287	0.059	0.267
<i>mrvr_ratio</i>	0.340	-0.339	-0.063	0.096	0.152	-0.204	-0.244
<i>total_fees_btc</i>	0.316	-0.033	0.081	-0.209	0.036	0.779	-0.472

Appendix B: Algorithms

Figure 23: K-Means PC1 vs PC2 Clusters

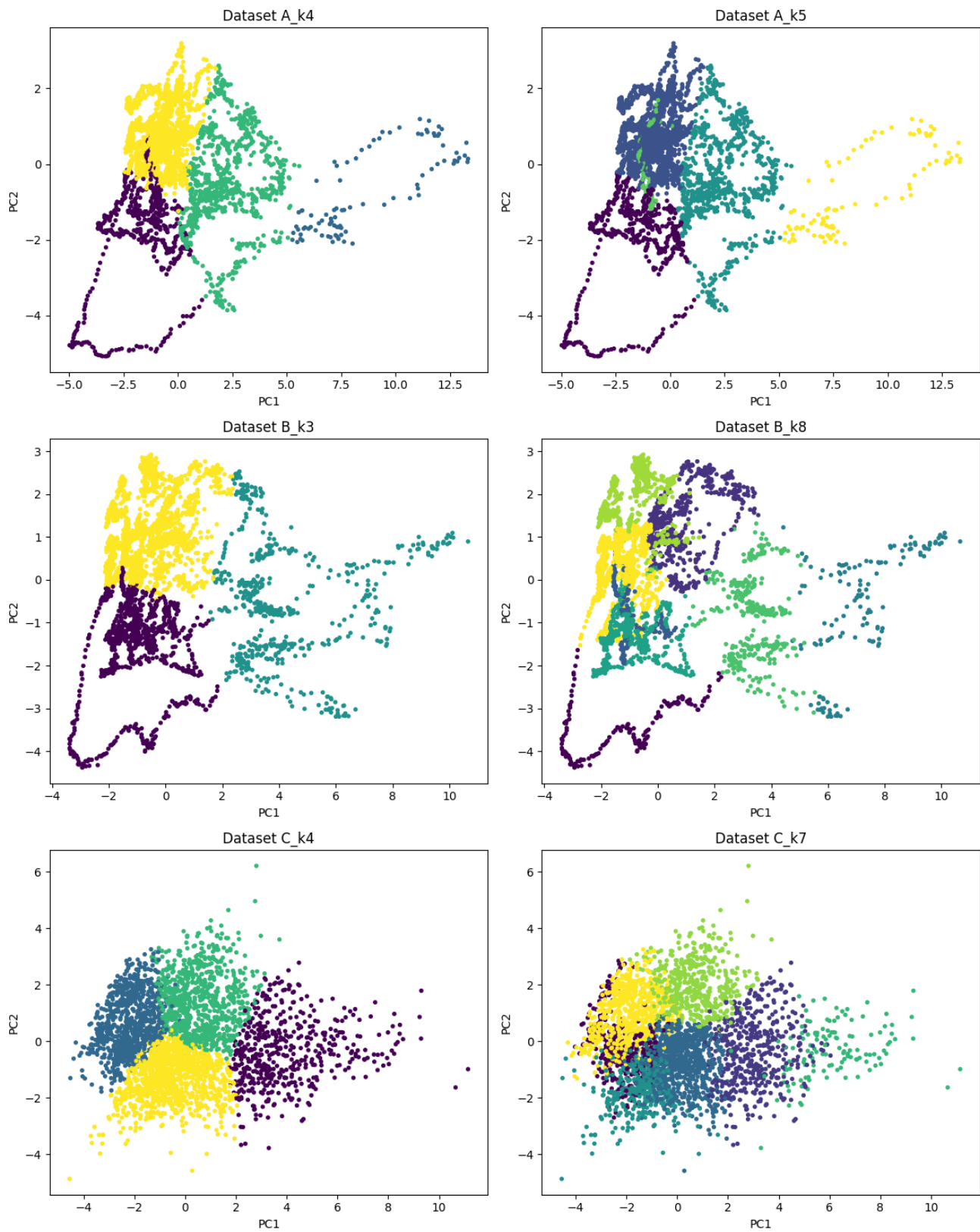


Figure 24: K-Means Clusters over Time

